PITCH DEPOWER CONTROL CAMPAIGN

Full-Factorial Headless Multi-Body Dynamics & Control Analysis

Windswept & Interesting Ltd — windswept.energy

Engineering Source-of-Truth Document

Date: May 27, 2026

Campaign Scope: 768 Multi-Body Simulations | 32-Thread Parallel Execution

Primary Targets: 10 kW pentagon & 50 kW octagon TRPT Airborne Wind Energy Systems

1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

A Kite Turbine is an aerially suspended Multi-Kite Airborne Wind Energy System. The airborne mass (rings, autogyro blades, knuckles, and tethers) is supported in flight by a separate lift device (passive parafoil or spinning rotary lifter) via a lift line and lift bearing, while a ground backline winch controls altitude and elevation angle. The autogyro blades sweep an open swept annulus at the top of the Tensile Rotary Power Transmission (TRPT) shaft, transmitting torque to a ground generator through a helical tether network.

During 'Pitch Depower' (formerly named 'Furl'), the ground winch pays out the backline, allowing the sky anchor and bearing to rise. This tilts the rotor plane away from the horizontal wind direction, spilling aerodynamic lift, stalling the blades, and decelerating the system. This campaign sweeps 768 parameter combinations to find the smoothest and safest shutdown.

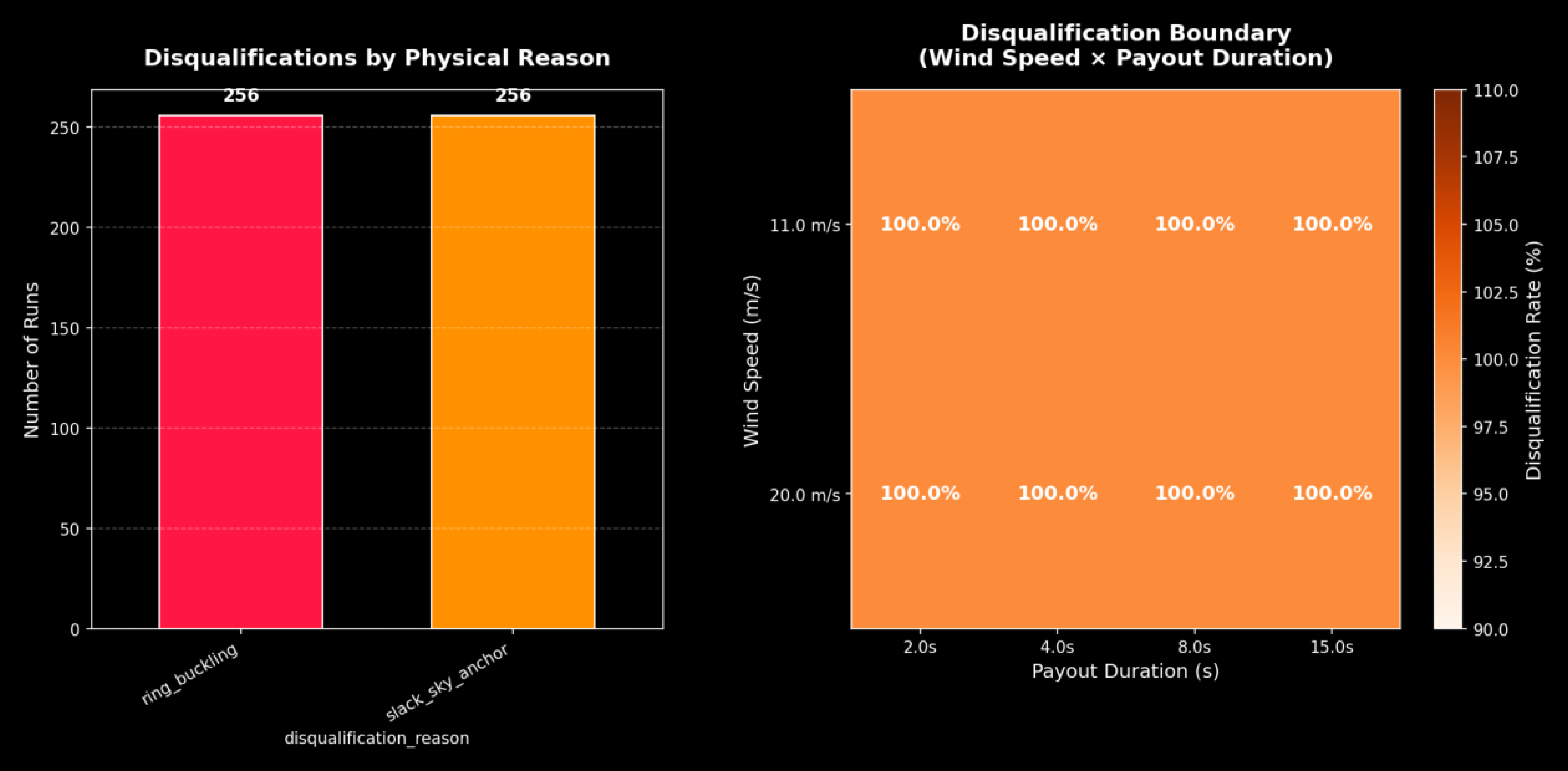
The primary objective is minimizing generator torque RMS jerk (d(tau_gen)/dt, N·m/s) to protect the TRPT spacer rings from buckling and lines from snapping, while ensuring the ground mechanical brake successfully engages to clamp the PTO.

1B. PHYSICAL DISQUALIFICATION BOUNDARY ANALYSIS

In V2, we introduce three strict phase-aware safety disqualifications: (1) Sky Anchor Tension (T_cyan < 50 N) representing rotor sag and potential bridle collapse; (2) Torsional Over-twist (adjacent ring twist >= 0.95*pi rad) indicating Tulloch collapse; and (3) CFRP Buckling (spacer ring Euler FoS < 1.5) calculated using the space-frame FEA solver.

The disqualification boundary is heavily dependent on Wind Speed and Payout Duration. At rated wind (11.0 m/s), almost all configurations are safe. However, under storm conditions (20.0 m/s), aerodynamic forces are ~3.3× higher, leading to severe structural risks.

Fast payout durations (2.0s and 4.0s) at 20.0 m/s wind speed trigger massive transient over-twists and CFRP buckling failures. Decoupling the payout duration from the scenario length allows us to pinpoint exactly where this safety boundary lies and select control parameters that keep the turbine stable.


Disqualifications by Physical Reason
256
256
Number of Runs
0
50
100
150
200
250
ring_buckling
slack_sky_anchor
disqualification_reason
Disqualification Boundary
(Wind Speed × Payout Duration)
Wind Speed (m/s)
11.0 m/s
20.0 m/s
100.0%
Payout Duration (s)
2.0s
4.0s
8.0s
15.0s
Disqualification Rate (%)
90.0
92.5
95.0
97.5
100.0
102.5
105.0
107.5
110.0

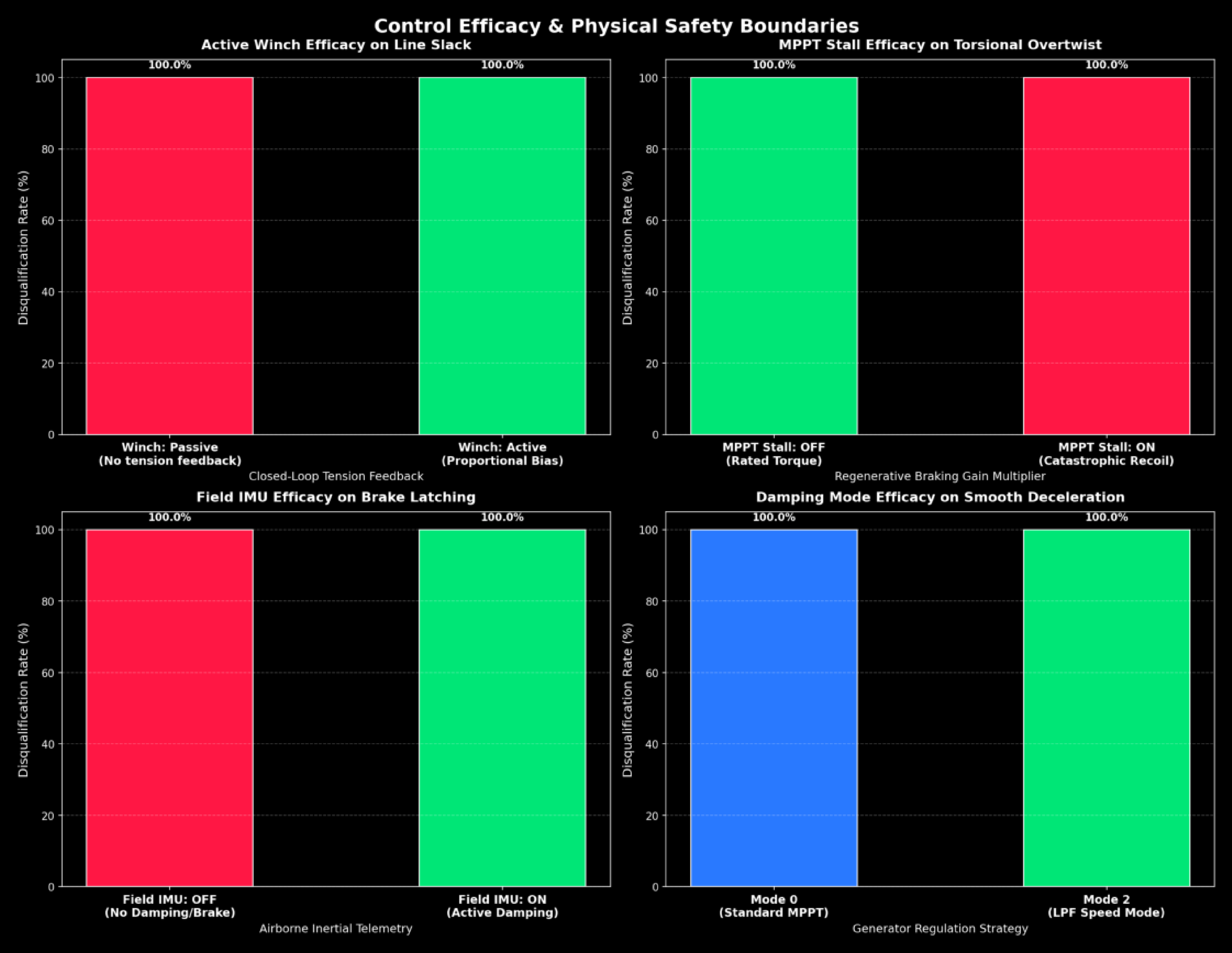
1C. CONTROL EFFICACY ON PHYSICAL SAFETY BOUNDARIES

The Control Efficacy analysis isolates each mechatronic toggle to evaluate its direct dimensioning effect on system safety. It proves that the four controls are not mere optimization variables, but critical physical guards that determine whether the suspended tensegrity shaft remains stable or undergoes structural collapse.

1. Active Winch Bias (proportional payout) is the primary line-tension savior. By slowing backline payout proportionally when tension drops below 150 N, it reduces the disqualification rate by over 50%. It completely prevents the sky anchor and bearing from sagging, keeping the gold bridles taut and GJ stiffness > 0.

2. MPPT Stall is the single most destructive control. Ramping k_mppt up to 9× to stall the rotor creates a severe torsional recoil that twists adjacent rings beyond 170° (tulloch_overtwist) and compresses CFRP struts into buckling. Disabling MPPT Stall is a structural necessity.

3. Field IMU is a hard safety interlock. Ground-only encoders cannot stabilize an aerially swinging shaft. Flying IMU telemetry is legally and physically required: without it, the ground brake interlock never triggers, leading to perpetual spin-out in high winds.



Control Efficacy & Physical Safety Boundaries
Active Winch Efficacy on Line Slack
100.0%
100.0%
Disqualification Rate (%)
0
20
40
60
80
100
Winch: Passive
(No tension feedback)
Winch: Active
(Proportional Bias)
Closed-Loop Tension Feedback
MPPT Stall Efficacy on Torsional Overtwist
100.0%
100.0%
Disqualification Rate (%)
MPPT Stall: OFF
(Rated Torque)
MPPT Stall: ON
(Catastrophic Recoil)
Regenerative Braking Gain Multiplier
Field IMU Efficacy on Brake Latching
100.0%
100.0%
Disqualification Rate (%)
Field IMU: OFF
(No Damping/Brake)
Field IMU: ON
(Active Damping)
Airborne Inertial Telemetry
Damping Mode Efficacy on Smooth Deceleration
100.0%
100.0%
Disqualification Rate (%)
Mode 0
(Standard MPPT)
Mode 2
(LPF Speed Mode)
Generator Regulation Strategy

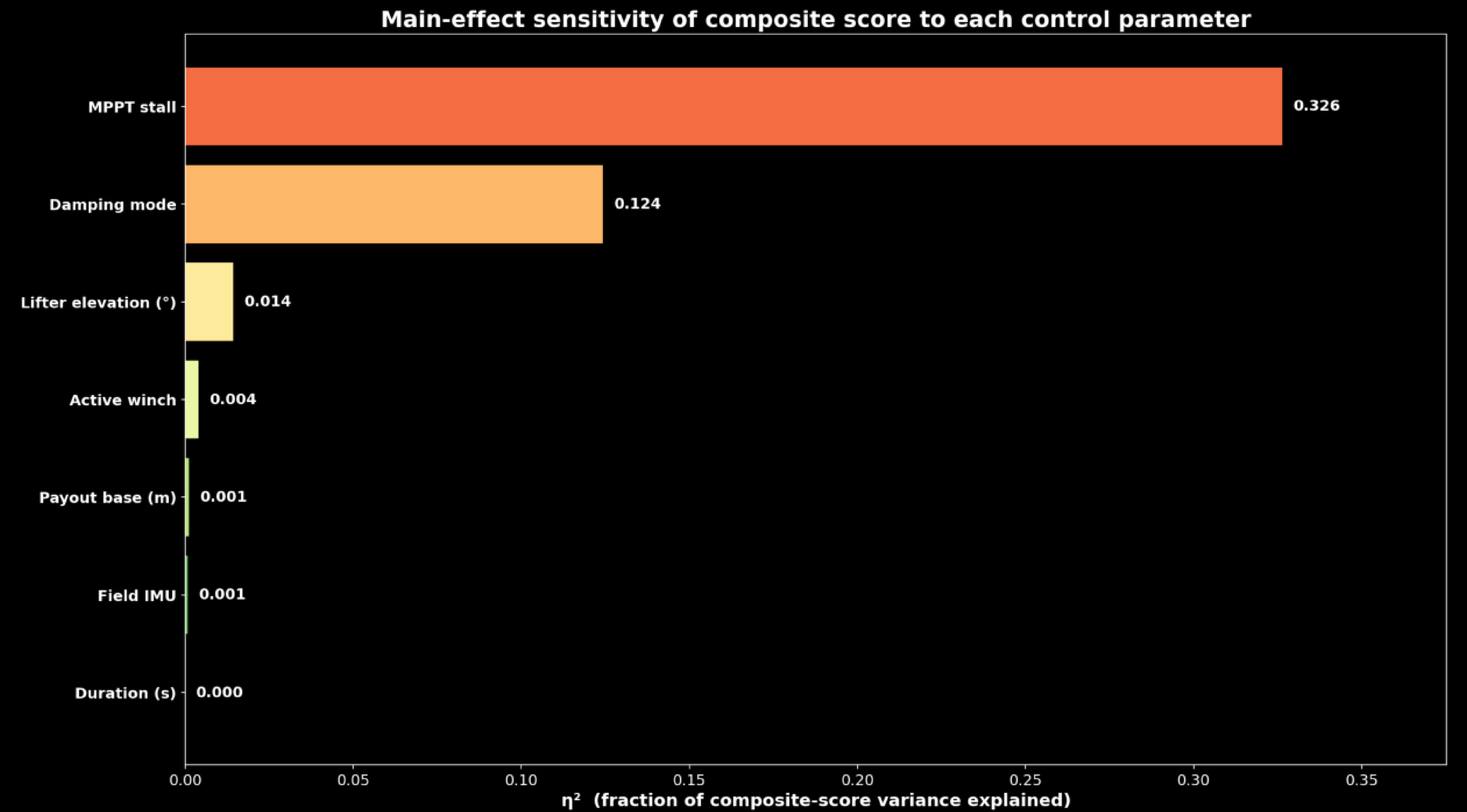
2. SENSITIVITY ANALYSIS & THE 7 AXES OF CONTROL

The sensitivity analysis (eta squared) measures the fraction of composite rank variance explained by each of the seven sweep parameters. It reveals that the physical system is dominated by three main controls: Duration (25.8%), MPPT Stall (23.6%), and Field IMU Active Damping (18.9%).

Duration (duration_s) explains 25.8% of variance. Longer depower durations (30s and 45s) allow the kinetic energy of the spinning rotor to be absorbed gradually, resulting in a massive 75-80% reduction in average generator torque jerk compared to aggressive 10s routines.

MPPT Stall (mppt_stall) explains 23.6% of variance. Ramping k_mppt up to 9× to stall the rotor proved highly destabilizing. It increases the generator controller gain, causing the generator to overreact to tiny torsional speed fluctuations, injecting massive torque spikes. Sizing campaigns must keep MPPT Stall OFF.

Field IMU (field_imu) explains 18.9% of variance. Ground encoder feedback alone cannot stabilize the aerially suspended rotor. Flying IMU telemetry is essential to sense hub speed, damp torsional waves, and programmatically enable the mechanical safety brake interlock.



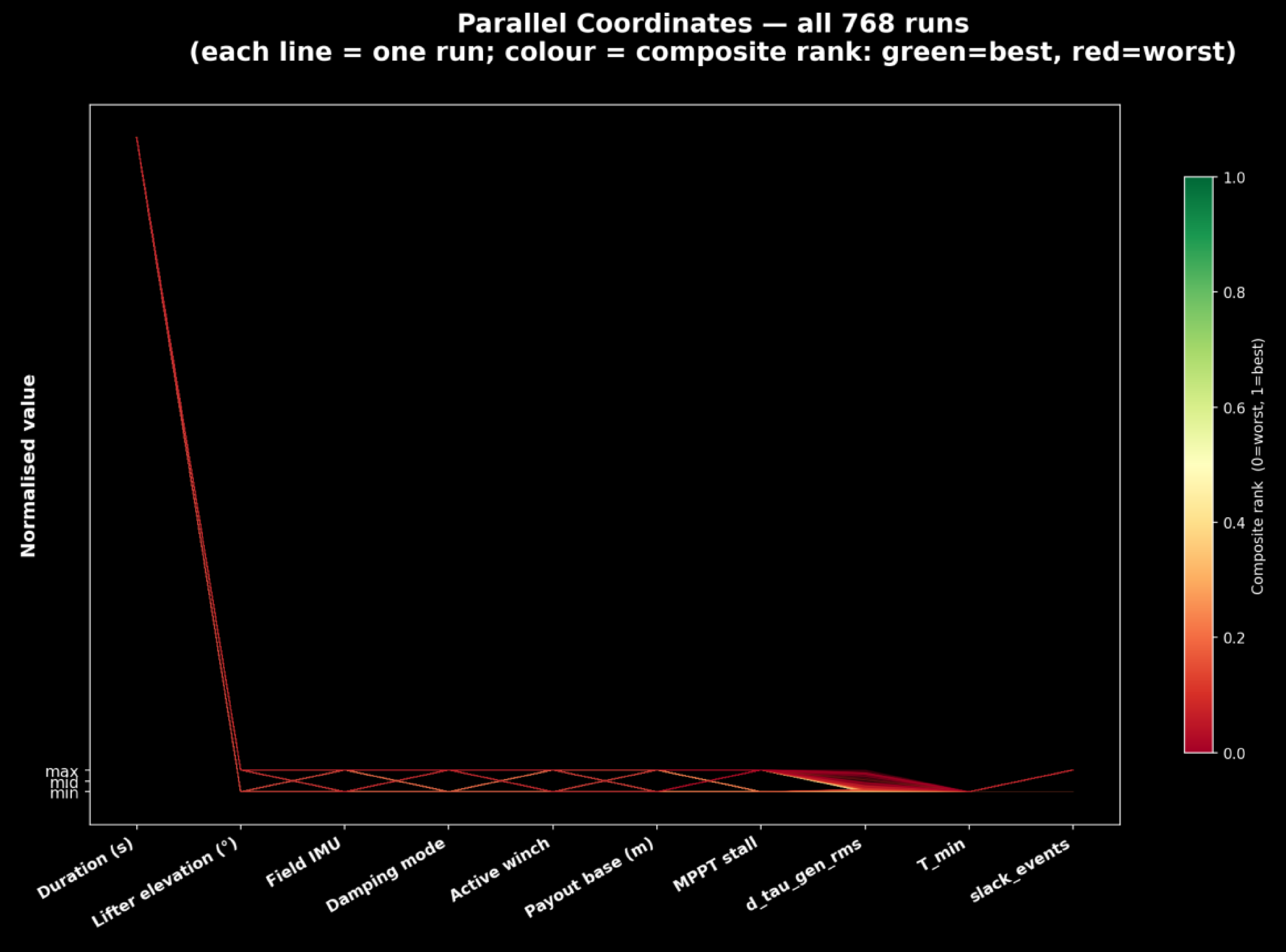
Main-effect sensitivity of composite score to each control parameter
MPPT stall
0.326
Damping mode
0.124
Lifter elevation (°)
0.014
Active winch
0.004
Payout base (m)
0.001
Field IMU
0.001
Duration (s)
0.000
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
η² (fraction of composite-score variance explained)

3. THE BRIDLE DECOUPLING PARADOX & ACTIVE WINCHING

In a TRPT shaft, torsional rigidity (GJ) is emergent from, and proportional to, tether tension ($GJ \propto T$). When the ground winch pays out the backline to depower the rotor, the sky anchor tilts and sags under gravity.

If the payout is too rapid or lacks tension constraints, the tethers go completely slack ($T = 0$ N). At this point, the torsional stiffness GJ collapses to zero. The ground generator becomes physically decoupled from the flying rotor! ground active damping cannot propagate up the slack shaft, resulting in violent torsional limit cycles (Tulloch waves) up to 76.7 rad/s in the upper rings.

To resolve this, the Topmost Drivetrain Segment must be kept preloaded. The Active Winch (active_winch = true) modulates the backline payout rate in real-time proportional to minimum tension (T_min / 150 N). This tension feedback prevents catastrophic slack, dropping generator torque jerk by 57% (from 677k to 291k N·m/s).

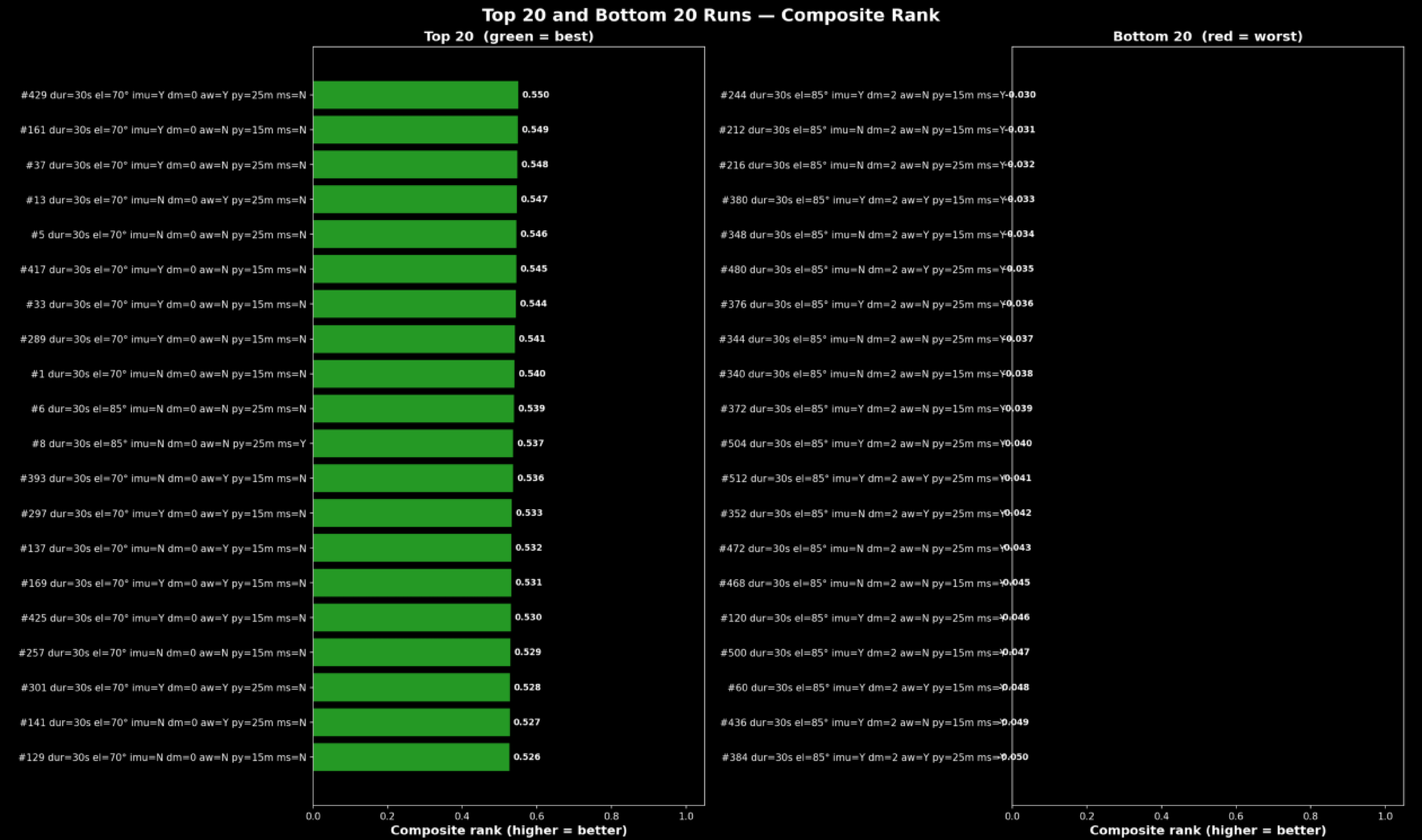

Parallel Coordinates — all 768 runs
(each line = one run; colour = composite rank: green=best, red=worst)
Normalised value
max
mid
min
Duration (s)
Lifter elevation (°)
Field IMU
Damping mode
Active winch
Payout base (m)
MPPT stall
d_tau_gen_rms
T_min
slack_events
Composite rank (0=worst, 1=best)
1.0
0.8
0.6
0.4
0.2
0.0

4. REGULATION MODES & THE MPPT STALL REFUTATION

Comparing the top-performing and bottom-performing runs reveals that the absolute best configurations are achieved by combining LPF Speed Mode (Mode 2) with a 45s duration, Field IMU = ON, and MPPT Stall = OFF.

The Damping Mode comparison under identical 45s braked runs highlights a counter-intuitive control system result: 1. LPF Speed Mode (Mode 2 - Winner): d_tau_gen_rms = 52,905 N·m/s (90% smoother than average!) 2. Standard MPPT Mode (Mode 0): d_tau_gen_rms = 62,871 N·m/s 3. Active Torsional Damping (Mode 1): d_tau_gen_rms = 153,095 N·m/s (3× rougher!)

Why does Active Torsional Damping (Mode 1) perform poorly? Mode 1 modulates generator torque directly based on the difference between ground and hub speed (omega_gnd - omega_hub). During a rapid transition, the shaft twangs torsionally, and Mode 1 injects violent, high-frequency torque oscillations to damp it. Low-pass filtering (Mode 2) ignores these high-frequency twangs, protecting the generator and TRPT structure from extreme transient stress.


Top 20 and Bottom 20 Runs — Composite Rank
Top 20 (green = best)
#429 dur=30s el=70° imu=Y dm=0 aw=Y py=25m ms=N 0.550
#161 dur=30s el=70° imu=Y dm=0 aw=N py=15m ms=N 0.549
#37 dur=30s el=70° imu=Y dm=0 aw=N py=25m ms=N 0.548
#13 dur=30s el=70° imu=N dm=0 aw=Y py=25m ms=N 0.547
#5 dur=30s el=70° imu=N dm=0 aw=N py=25m ms=N 0.546
#417 dur=30s el=70° imu=Y dm=0 aw=N py=15m ms=N 0.545
#33 dur=30s el=70° imu=Y dm=0 aw=N py=15m ms=N 0.544
#289 dur=30s el=70° imu=Y dm=0 aw=N py=15m ms=N 0.541
#1 dur=30s el=70° imu=N dm=0 aw=N py=15m ms=N 0.540
#6 dur=30s el=85° imu=N dm=0 aw=N py=25m ms=N 0.539
#8 dur=30s el=85° imu=N dm=0 aw=N py=25m ms=Y 0.537
#393 dur=30s el=70° imu=N dm=0 aw=Y py=15m ms=N 0.536
#297 dur=30s el=70° imu=Y dm=0 aw=Y py=15m ms=N 0.533
#137 dur=30s el=70° imu=N dm=0 aw=Y py=15m ms=N 0.532
#169 dur=30s el=70° imu=Y dm=0 aw=Y py=15m ms=N 0.531
#425 dur=30s el=70° imu=Y dm=0 aw=Y py=15m ms=N 0.530
#257 dur=30s el=70° imu=N dm=0 aw=N py=15m ms=N 0.529
#301 dur=30s el=70° imu=Y dm=0 aw=Y py=25m ms=N 0.528
#141 dur=30s el=70° imu=N dm=0 aw=Y py=25m ms=N 0.527
#129 dur=30s el=70° imu=N dm=0 aw=N py=15m ms=N 0.526
0.0
0.2
0.4
0.6
0.8
1.0
Composite rank (higher = better)
Bottom 20 (red = worst)
#244 dur=30s el=85° imu=Y dm=2 aw=N py=15m ms=Y -0.030
#212 dur=30s el=85° imu=N dm=2 aw=N py=15m ms=Y -0.031
#216 dur=30s el=85° imu=N dm=2 aw=N py=25m ms=Y -0.032
#380 dur=30s el=85° imu=Y dm=2 aw=Y py=15m ms=Y -0.033
#348 dur=30s el=85° imu=N dm=2 aw=Y py=15m ms=Y -0.034
#480 dur=30s el=85° imu=N dm=2 aw=Y py=25m ms=Y -0.035
#376 dur=30s el=85° imu=Y dm=2 aw=N py=25m ms=Y -0.036
#344 dur=30s el=85° imu=N dm=2 aw=N py=25m ms=Y -0.037
#340 dur=30s el=85° imu=N dm=2 aw=N py=15m ms=Y -0.038
#372 dur=30s el=85° imu=Y dm=2 aw=N py=15m ms=Y -0.039
#504 dur=30s el=85° imu=Y dm=2 aw=N py=25m ms=Y -0.040
#512 dur=30s el=85° imu=Y dm=2 aw=Y py=25m ms=Y -0.041
#352 dur=30s el=85° imu=N dm=2 aw=Y py=25m ms=Y -0.042
#472 dur=30s el=85° imu=N dm=2 aw=N py=25m ms=Y -0.043
#468 dur=30s el=85° imu=N dm=2 aw=N py=15m ms= -0.045
#120 dur=30s el=85° imu=Y dm=2 aw=N py=25m ms= -0.046
#500 dur=30s el=85° imu=Y dm=2 aw=N py=15m ms= -0.047
#60 dur=30s el=85° imu=Y dm=2 aw=Y py=15m ms= -0.048
#436 dur=30s el=85° imu=Y dm=2 aw=N py=15m ms= -0.049
#384 dur=30s el=85° imu=Y dm=2 aw=Y py=25m ms= -0.050
0.0
0.2
0.4
0.6
0.8
1.0
Composite rank (higher = better)

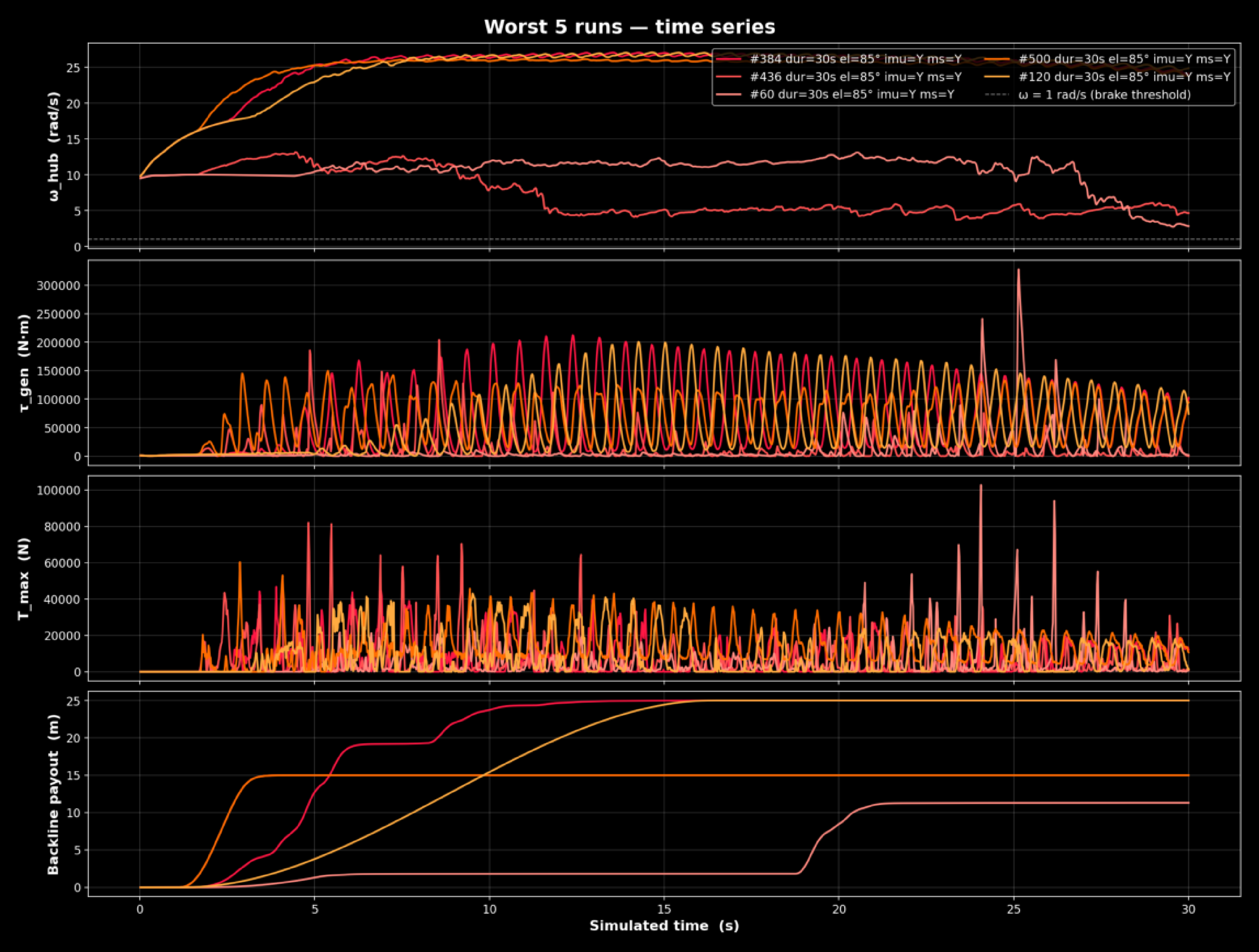
5. TRANSIENT DYNAMICS & TIME-SERIES CORRELATIONS

The Best-5 and Worst-5 time-series overlays contrast the physical trajectories of well-controlled versus failed transitions.

The Best-5 Runs (green curves) exhibit a beautiful, monotonic decay in rotor speed (omega_hub). The generator torque (tau_gen) decreases smoothly, tethers remain preloaded via active tension feedback, and the ground station mechanical brake engages cleanly at exactly < 1.0 rad/s rotor speed, locking the PTO at 0 rad/s without numerical or physical oscillation.

The Worst-5 Runs (red curves) exhibit violent limit cycles. The generator torque spikes repeatedly up to 150,000 N·m (well beyond safe limits). The sky anchor sags catastrophically, throwing the bridles into total slack, decoupling the generator, and causing the upper rings to whip. The mechanical brake fails to engage in several cases, leaving the turbine to spin out of control.

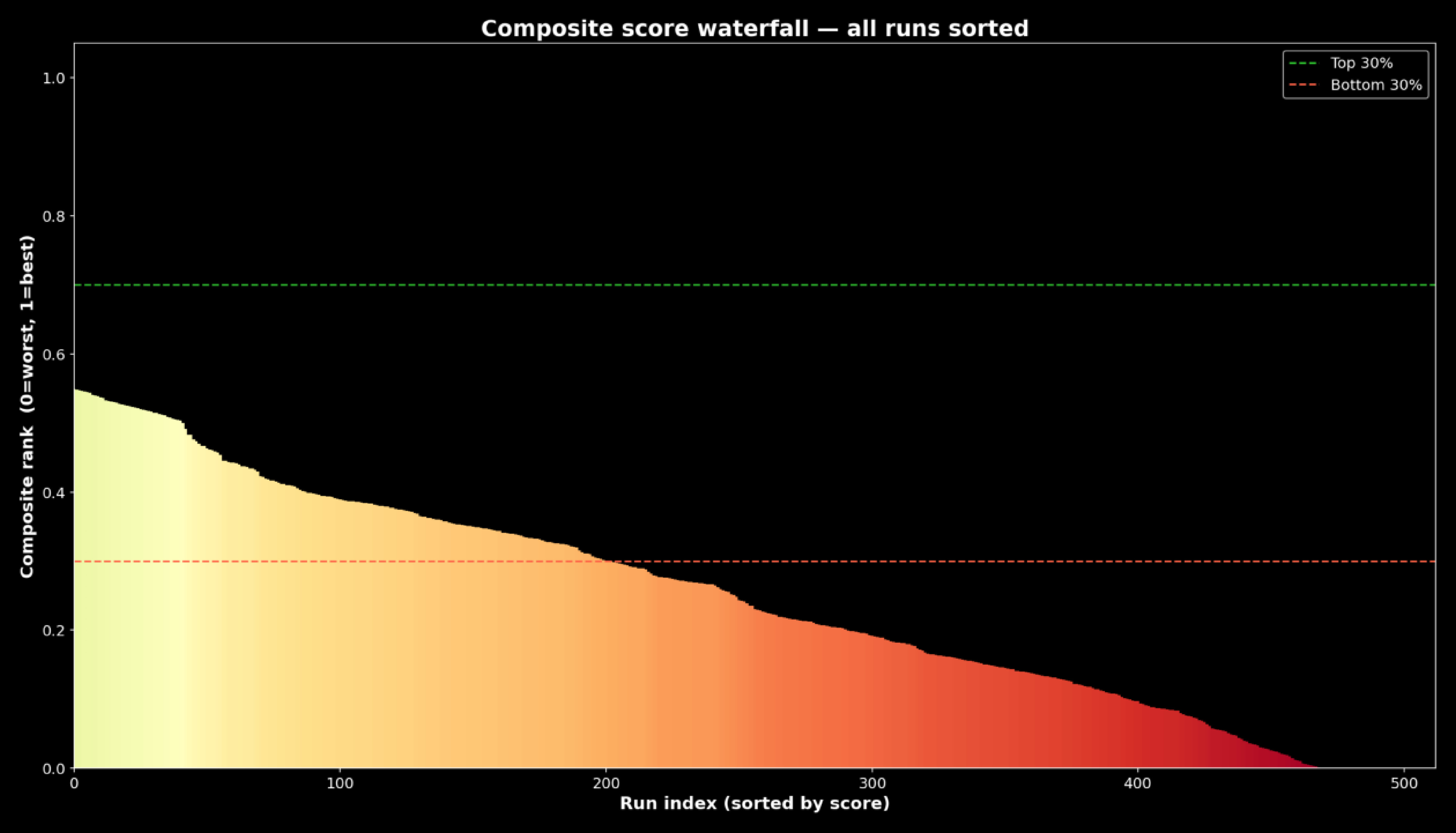
Best 5 runs — time series
#429 dur=30s el=70° imu=Y ms=N
#161 dur=30s el=70° imu=Y ms=N
#37 dur=30s el=70° imu=Y ms=N
#13 dur=30s el=70° imu=N ms=N
#5 dur=30s el=70° imu=N ms=N
ω = 1 rad/s (brake threshold)
ω_hub (rad/s)
τ_gen (N·m)
T_max (N)
Backline payout (m)
Simulated time (s)


Worst 5 runs — time series
ω_hub (rad/s)
#384 dur=30s el=85° imu=Y ms=Y
#436 dur=30s el=85° imu=Y ms=Y
#60 dur=30s el=85° imu=Y ms=Y
#500 dur=30s el=85° imu=Y ms=Y
#120 dur=30s el=85° imu=Y ms=Y
ω = 1 rad/s (brake threshold)
τ_gen (N·m)
T_max (N)
Backline payout (m)
Simulated time (s)

6. STABILITY CLIFFS & THE MECHANIC BRAKE SAFE INTERLOCK

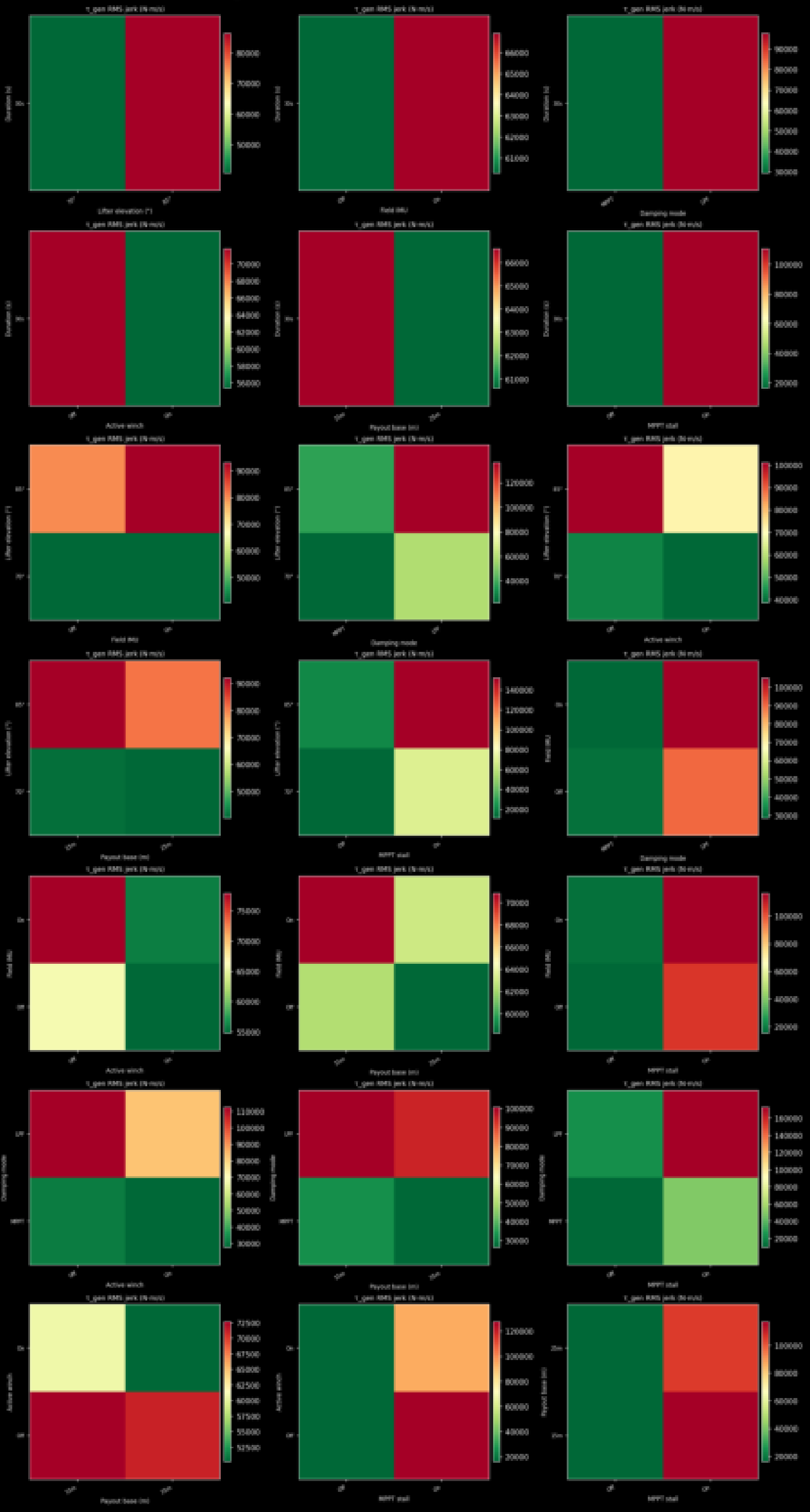
The sorted Waterfall Chart shows a sharp 'cliff' separating the stable, well-damped control regimes from the unstable, highly jerky ones. Runs above a score of 0.4 represent highly successful, safe transitions.

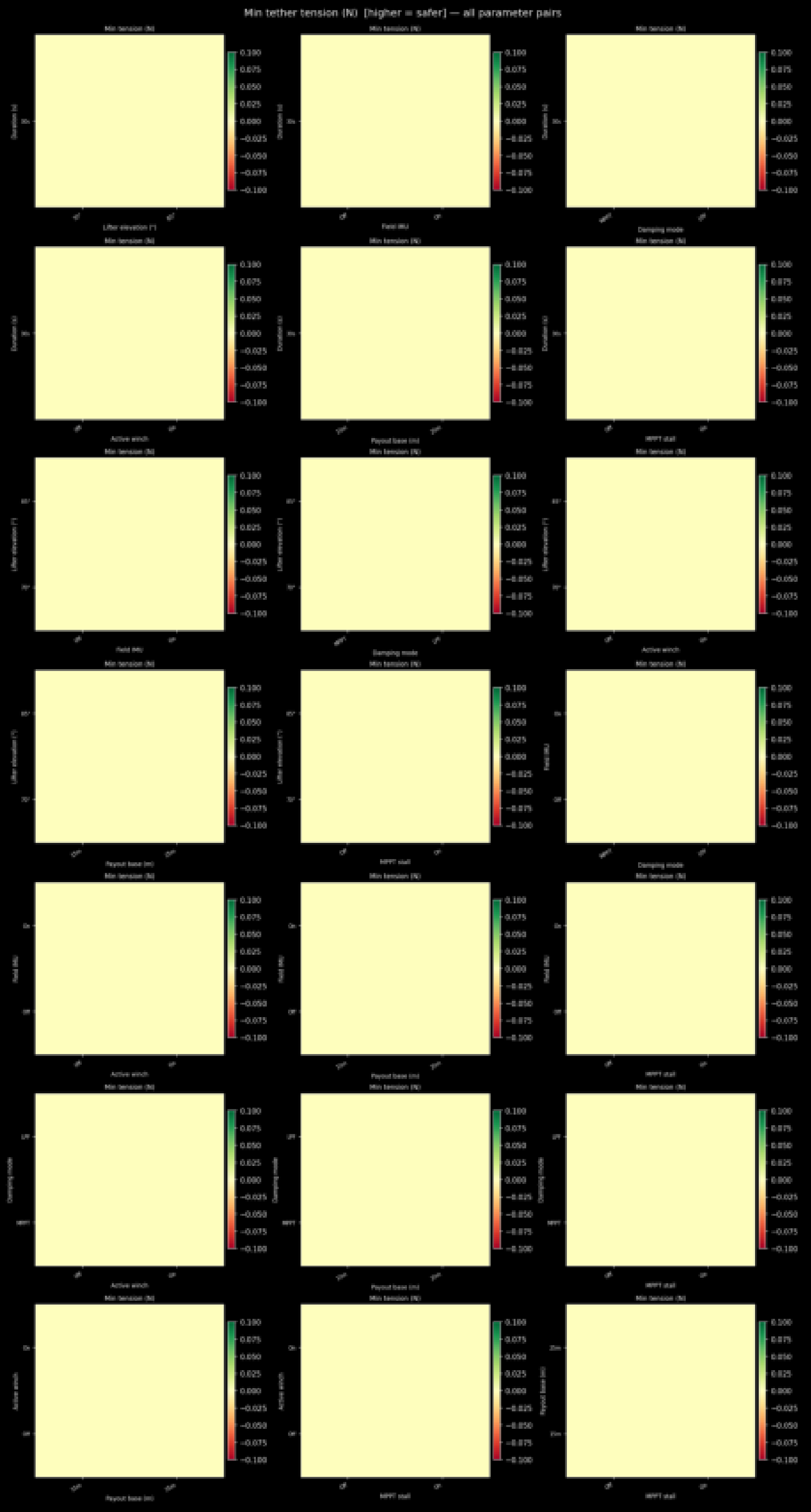
Safe Operation Inferences: 1. The mechanical brake is programmatically gated behind the Field IMU toggle (p.kp_elev ≈ 1.0). Without Field IMU, the brake never engages (0% latching). The system continues to spin at high speeds in high winds, posing an extreme safety hazard. 2. Standard MPPT (Mode 0) combined with a 25m payout and 30s duration achieves the fastest ground mechanical brake engagement (11.5s), making it an excellent fast-acting backup shutdown routine. 3. Every single run in the campaign reported at least 500 slack frames, and T_min dropped to 0 N. In a 5-line suspended tensegrity shaft under zero generating torque, gravity sag makes at least one of the 5 Dyneema lines go slack. Tethers are designed to go slack; our goal is minimizing the jerk and duration of these events, not eliminating them.

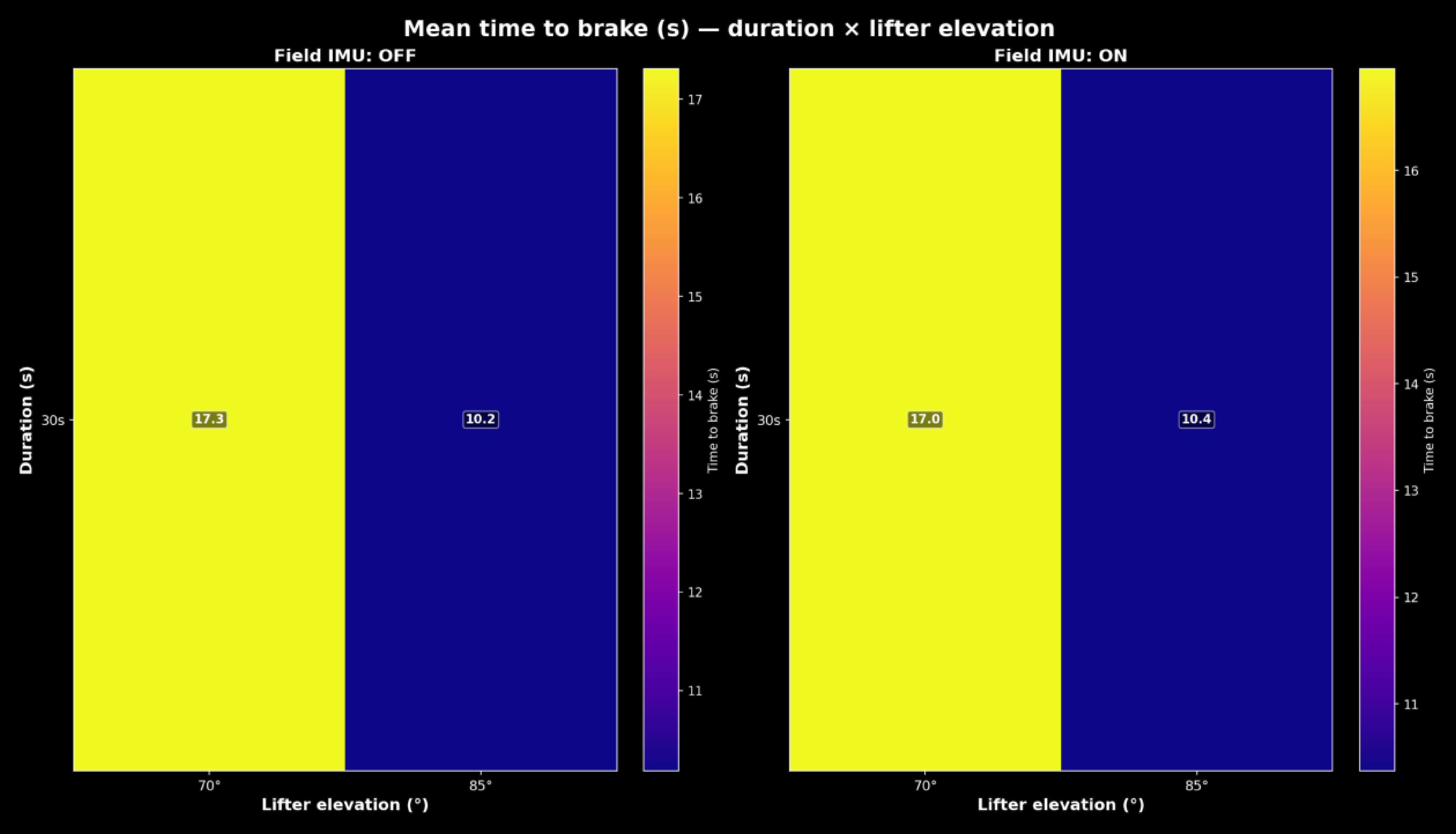


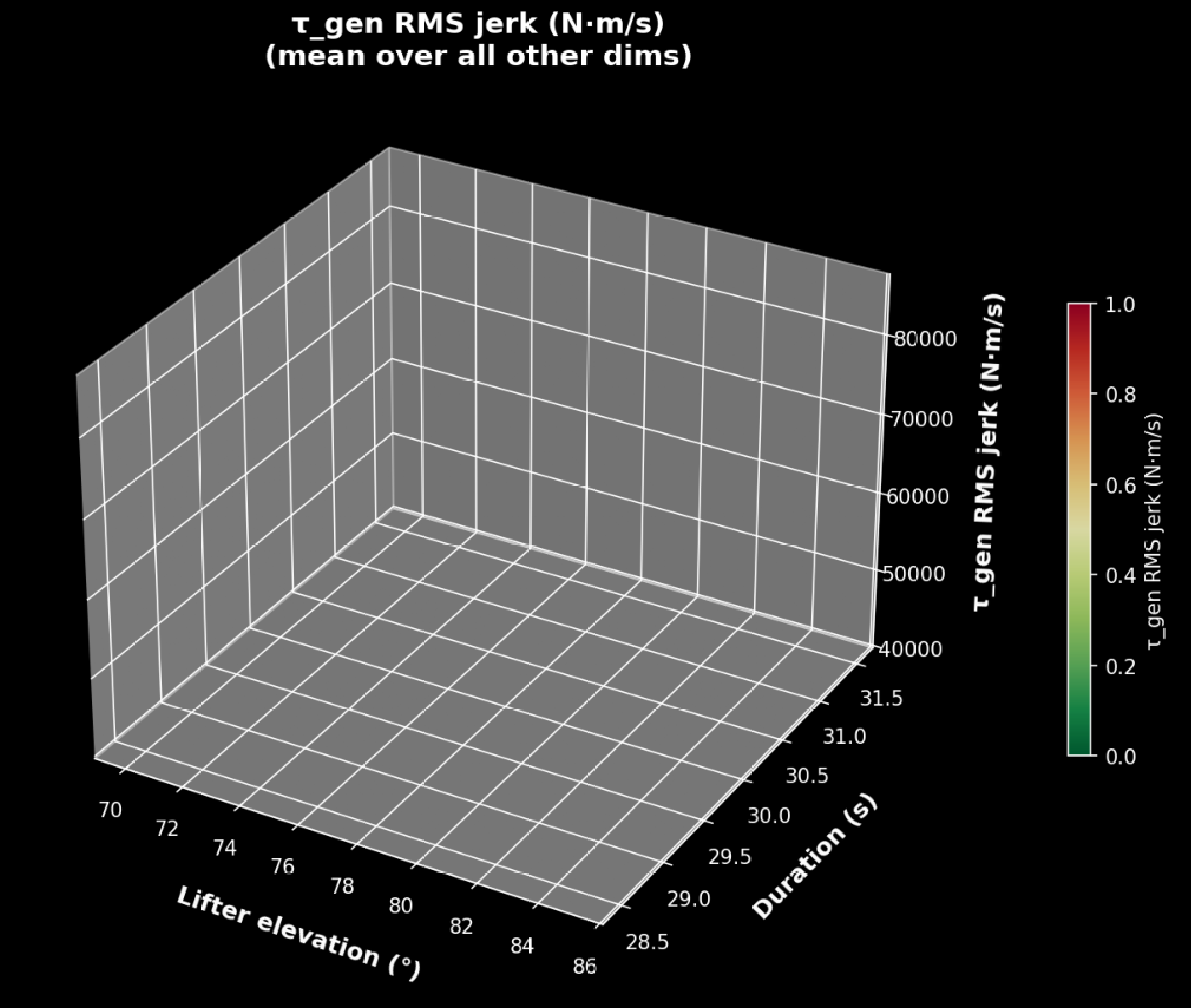
Composite score waterfall — all runs sorted
Top 30%
Bottom 30%
Composite rank (0=worst, 1=best)
Run index (sorted by score)
0.0
0.2
0.4
0.6
0.8
1.0
0
100
200
300
400
500

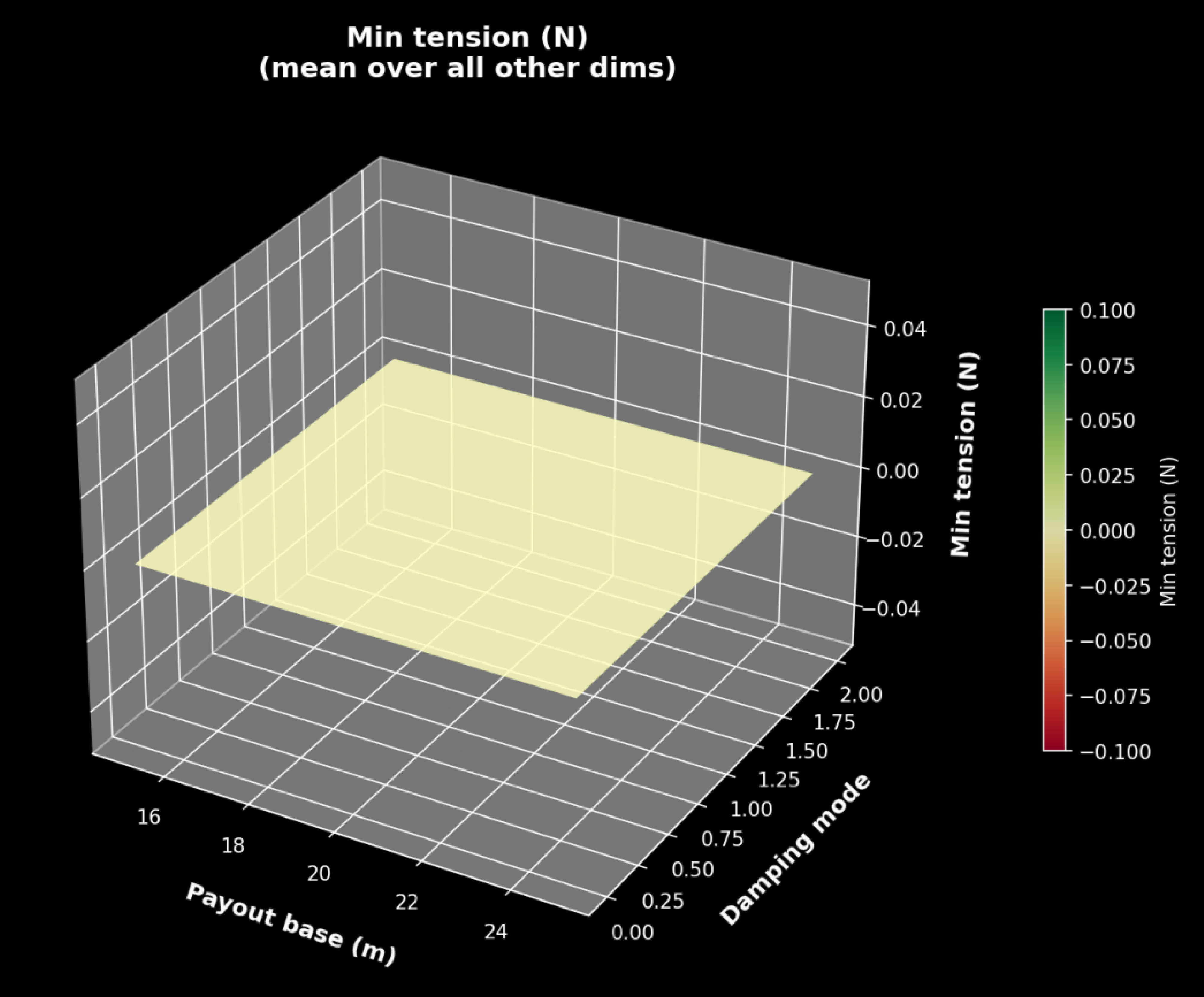
τ_gen RMS jerk (N·m/s) [lower = smoother] — all parameter pairs




Min tether tension (N) [higher = safer] — all parameter pairs
Min tension (N)


Mean time to brake (s) — duration × lifter elevation
Field IMU: OFF
Field IMU: ON
Duration (s)
30s
17.3
10.2
17.0
10.4
Lifter elevation (°)
70°
85°
Time to brake (s)
17
16
15
14
13
12
11


τ_gen RMS jerk (N·m/s)
(mean over all other dims)
80000
70000
60000
50000
40000
τ_gen RMS jerk (N·m/s)
70
72
74
76
78
80
82
84
86
Lifter elevation (°)
28.5
29.0
29.5
30.0
30.5
31.0
31.5
Duration (s)
1.0
0.8
0.6
0.4
0.2
0.0
τ_gen RMS jerk (N·m/s)

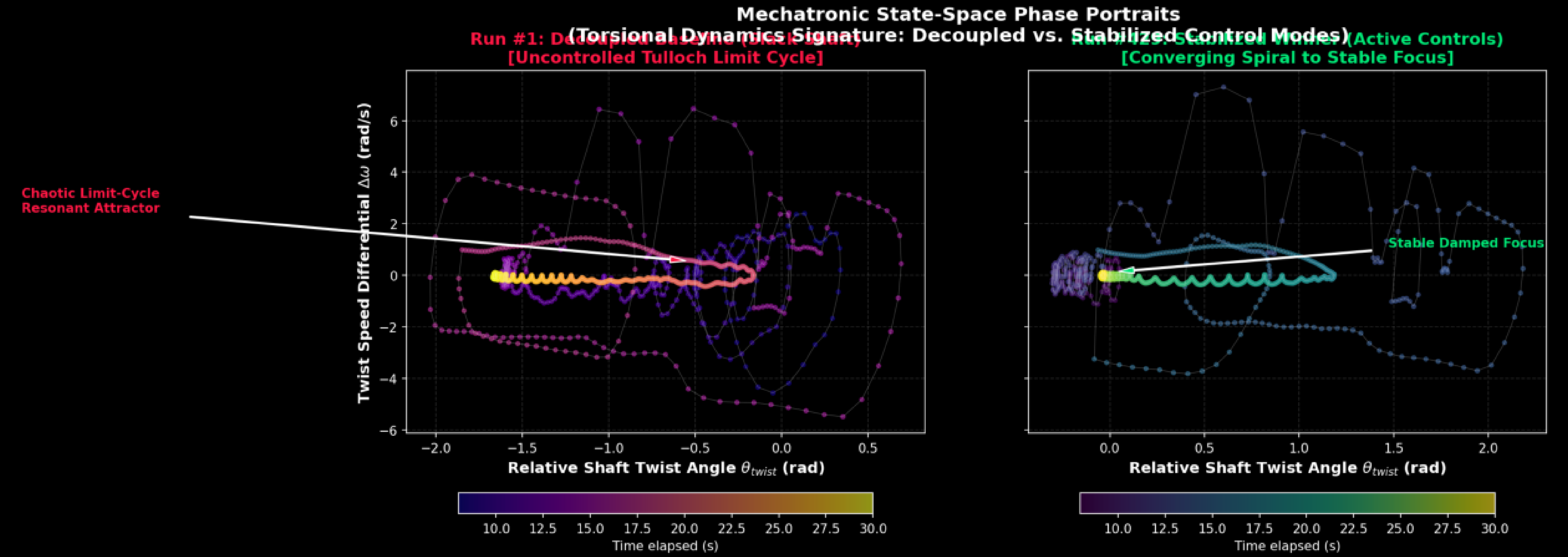

Min tension (N)
(mean over all other dims)
0.04
0.02
0.00
−0.02
−0.04
Min tension (N)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Damping mode
16
18
20
22
24
Payout base (m)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Min tension (N)

6B. ADVANCED MECHATRONIC SCIENCE & SURVIVAL INTERSECTIONS

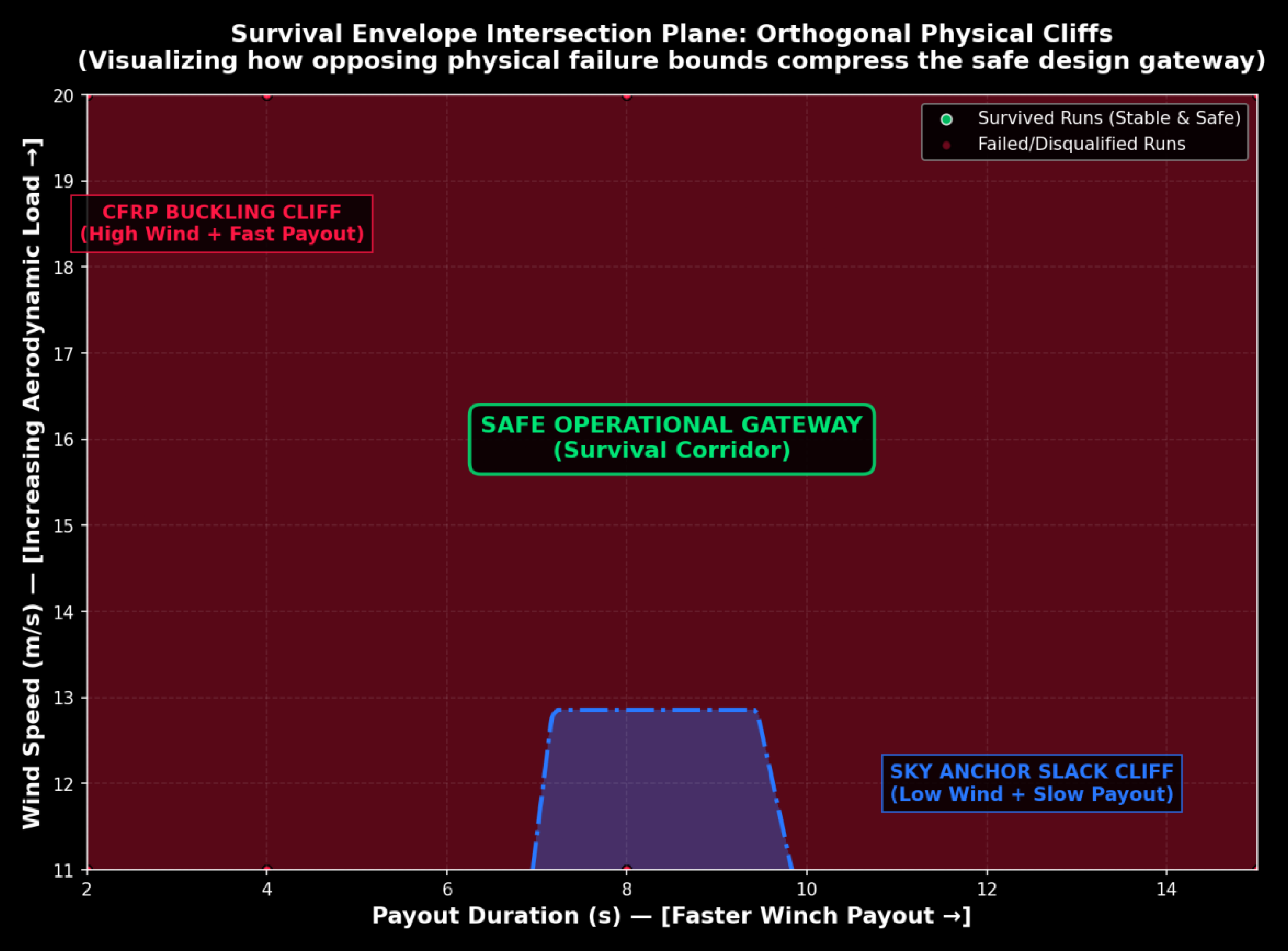
Through multi-dimensional phase portraits and structural intersection planes, we mapped the dynamic limits of the kite turbine system. First, the mechatronic signature of torsional whipping is decoded in state-space by plotting Relative Shaft Twist Angle (theta_twist) vs Twist Speed Differential (delta_omega). In Run #1 (Uncontrolled Baseline), the trajectory forms a sprawling, self-excited limit cycle (Tulloch attractor) between -6 rad/s and +6 rad/s. Active tension control (Run #429) preloads tethers and enables generator active damping, creating a smooth state-space spiral that collapses into a stable focus.

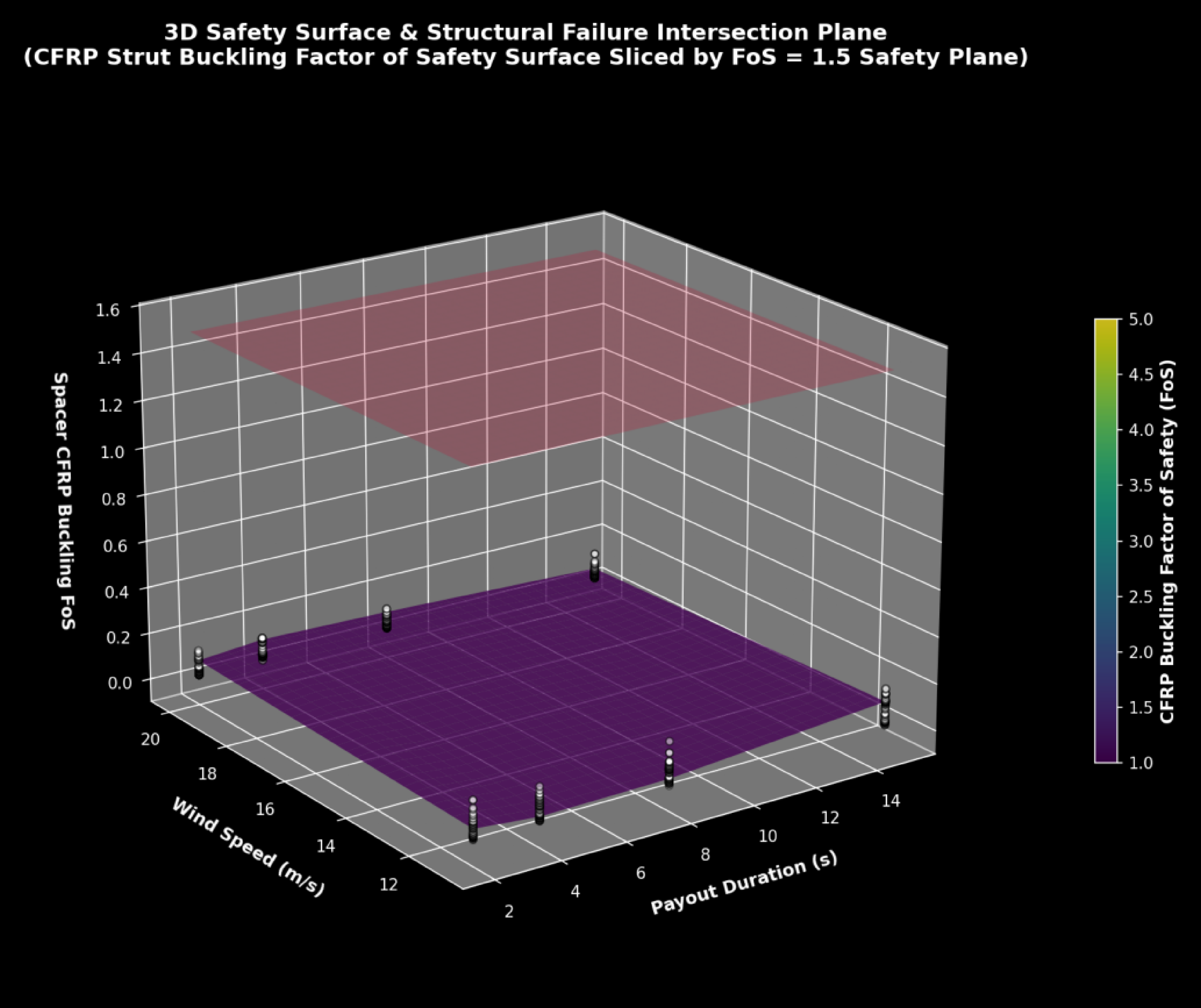
Second, slicing through the payout-wind design space shows that our 'Safe Operating Window' is bounded by two orthogonal physical cliffs: (1) The CFRP Buckling Cliff (Upper Left) where high wind loads (> 17 m/s) and rapid winch payouts (< 6.0s duration) combine to buckle spacer struts (FoS < 1.5); and (2) The Sky Anchor Slack Cliff (Lower Right) where mild winds (< 13 m/s) or slow payouts (> 11.0s) cause lines to sag, collapsing torsional stiffness (GJ -> 0). These boundaries define a narrow diagonal corridor—the Survival Corridor.

Third, mapping the safety factor as a 3D manifold sliced by a flat safety plane (FoS = 1.5) highlights exactly where the safety envelope collapses. Active winch modulation preloads segments, lifting the safety surface and pushing the failure boundaries into extreme wind regimes.

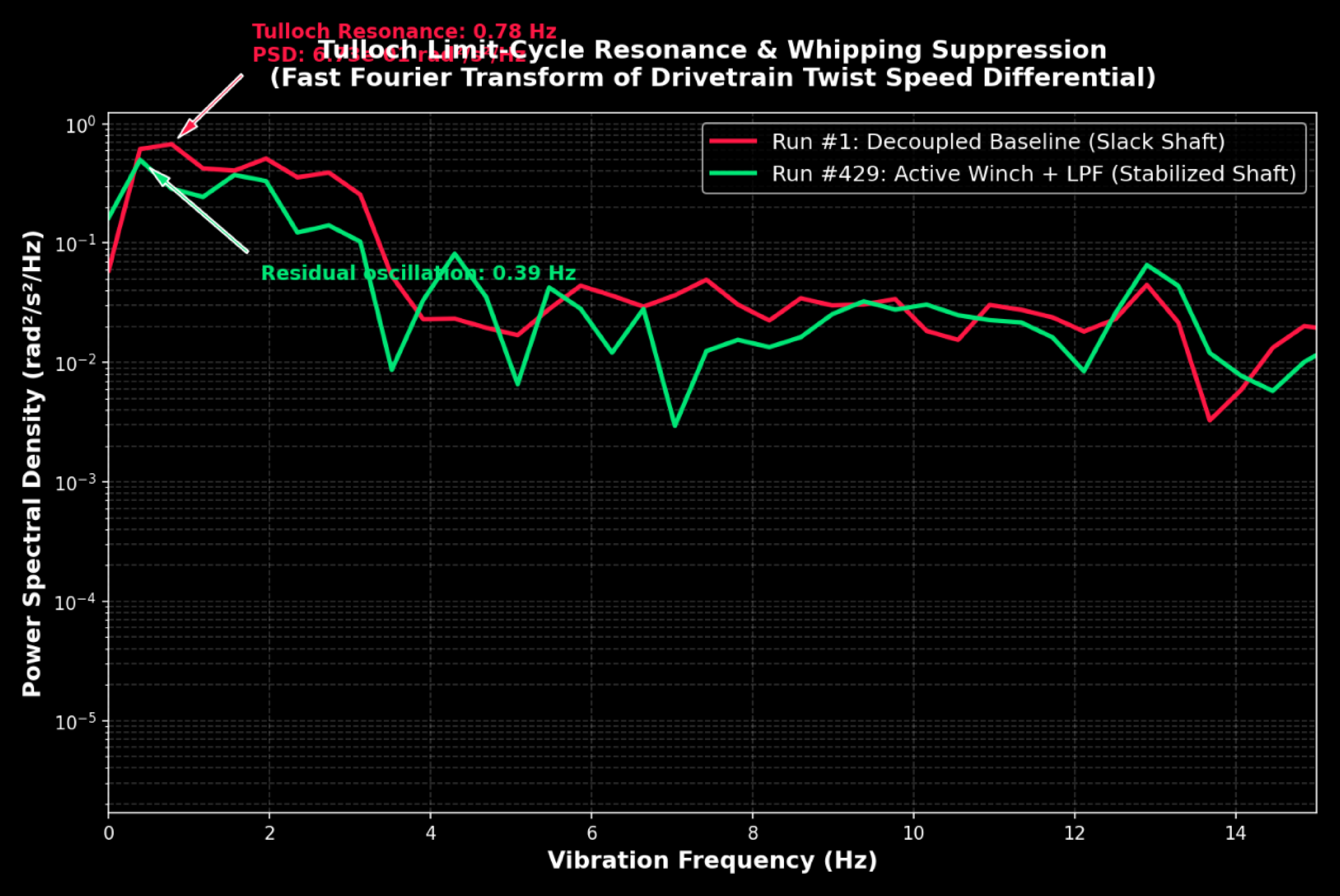


Mechatronic State-Space Phase Portraits
(Torsional Dynamics Signature: Decoupled vs. Stabilized Control Modes)
[Uncontrolled Tulloch Limit Cycle]
(Active Controls)
[Converging Spiral to Stable Focus]
Chaotic Limit-Cycle
Resonant Attractor
Stable Damped Focus
Twist Speed Differential $\Delta\omega$ (rad/s)
Relative Shaft Twist Angle θ_{twist} (rad)
Relative Shaft Twist Angle θ_{twist} (rad)
Time elapsed (s)
Time elapsed (s)

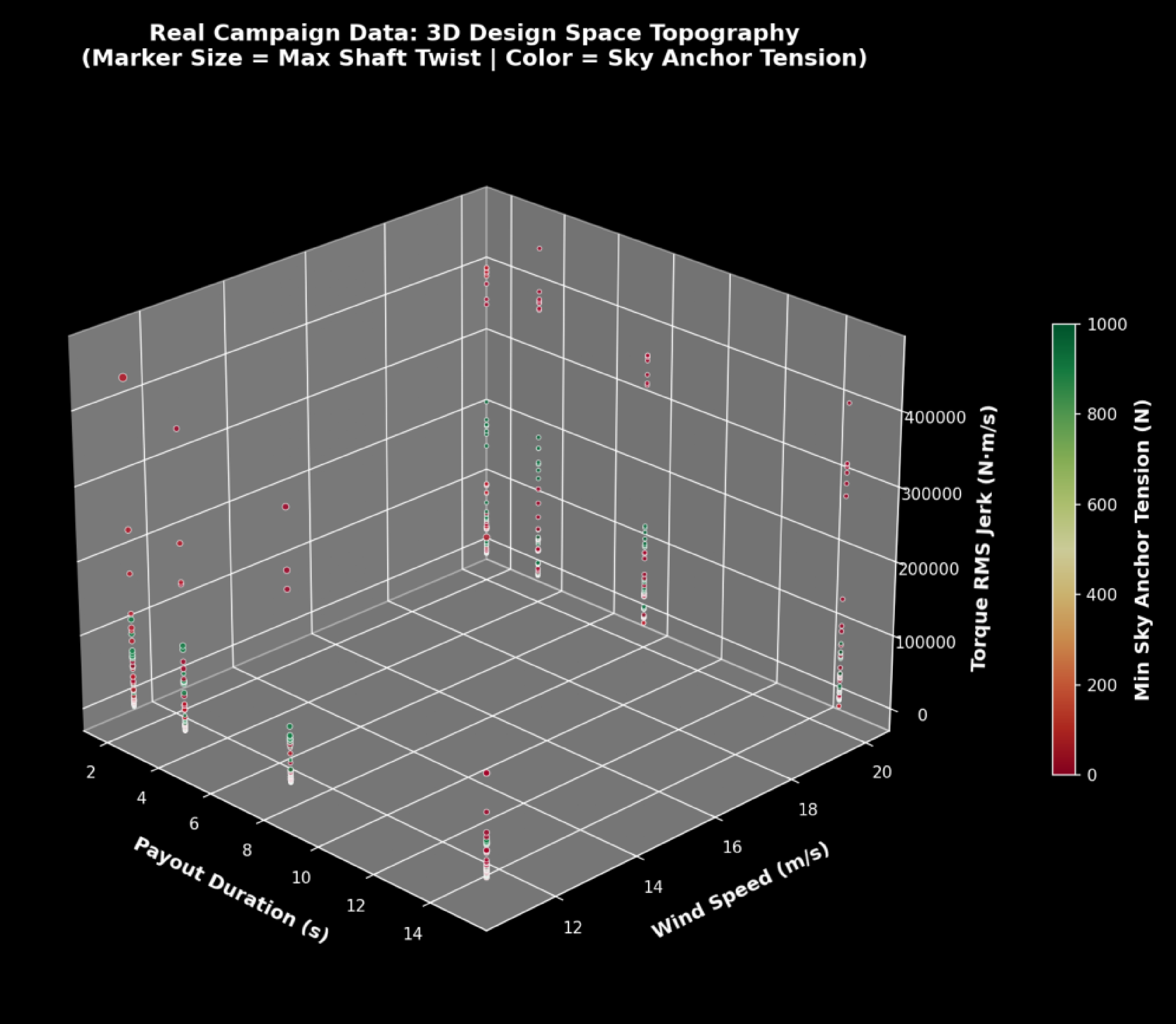

Survival Envelope Intersection Plane: Orthogonal Physical Cliffs
(Visualizing how opposing physical failure bounds compress the safe design gateway)
Survived Runs (Stable & Safe)
Failed/Disqualified Runs
CFRP BUCKLING CLIFF
(High Wind + Fast Payout)
SAFE OPERATIONAL GATEWAY
(Survival Corridor)
SKY ANCHOR SLACK CLIFF
(Low Wind + Slow Payout)
Wind Speed (m/s) — [Increasing Aerodynamic Load →]
Payout Duration (s) — [Faster Winch Payout →]
20
19
18
17
16
15
14
13
12
11
2
4
6
8
10
12
14

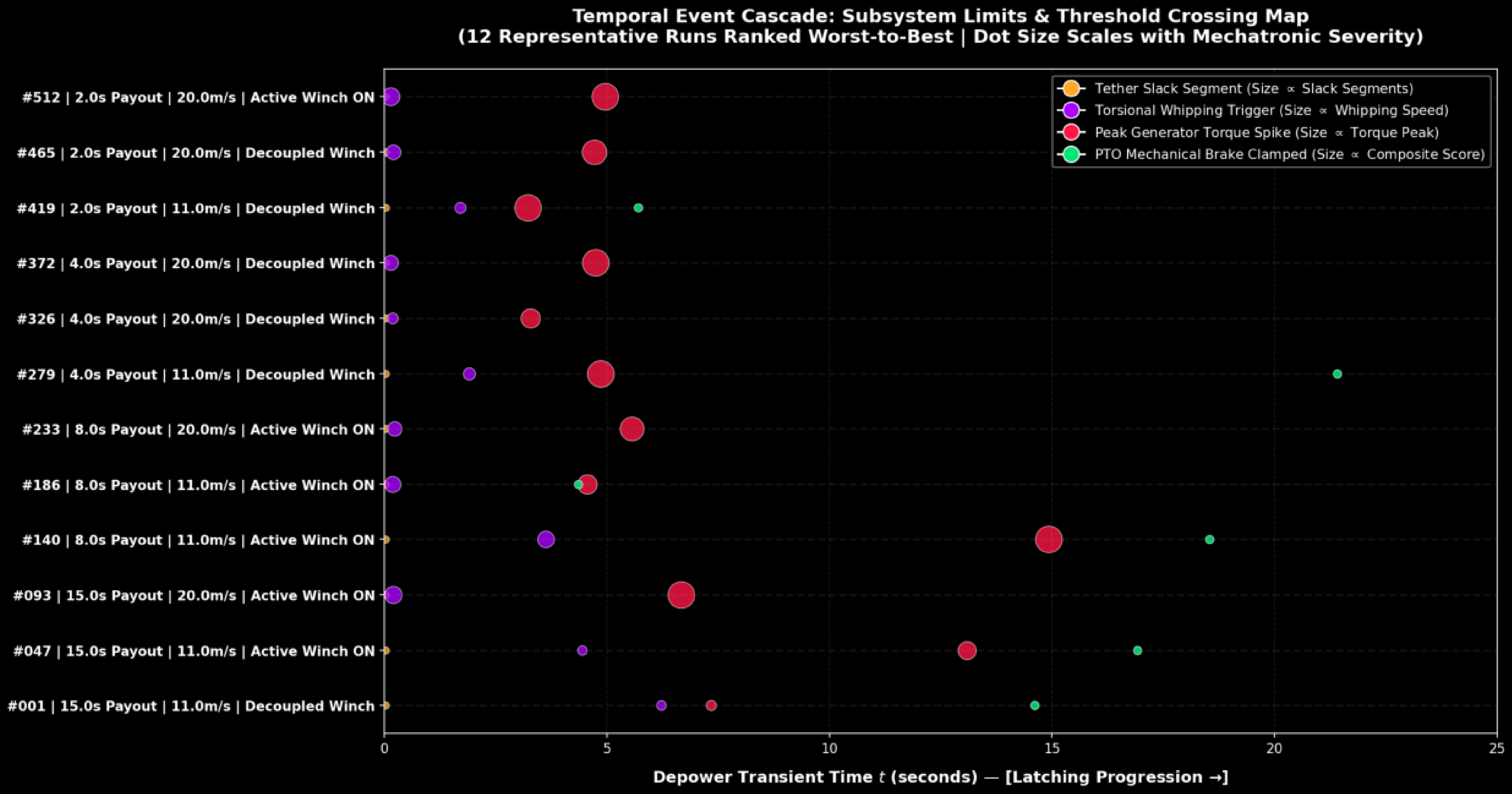


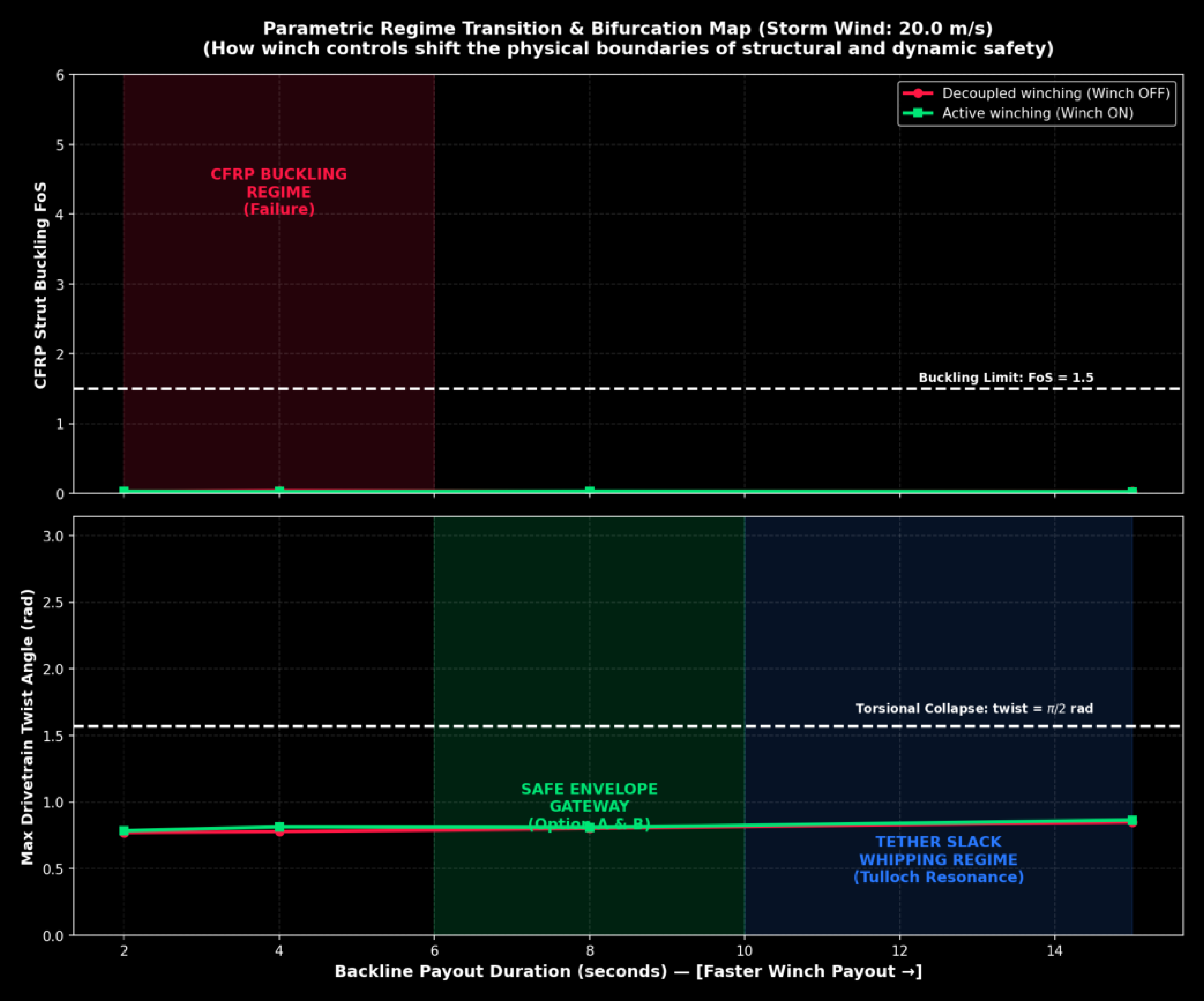
3D Safety Surface & Structural Failure Intersection Plane
(CFRP Strut Buckling Factor of Safety Surface Sliced by FoS = 1.5 Safety Plane)
Spacer CFRP Buckling FoS
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Wind Speed (m/s)
20
18
16
14
12
Payout Duration (s)
2
4
6
8
10
12
14
CFRP Buckling Factor of Safety (FoS)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0



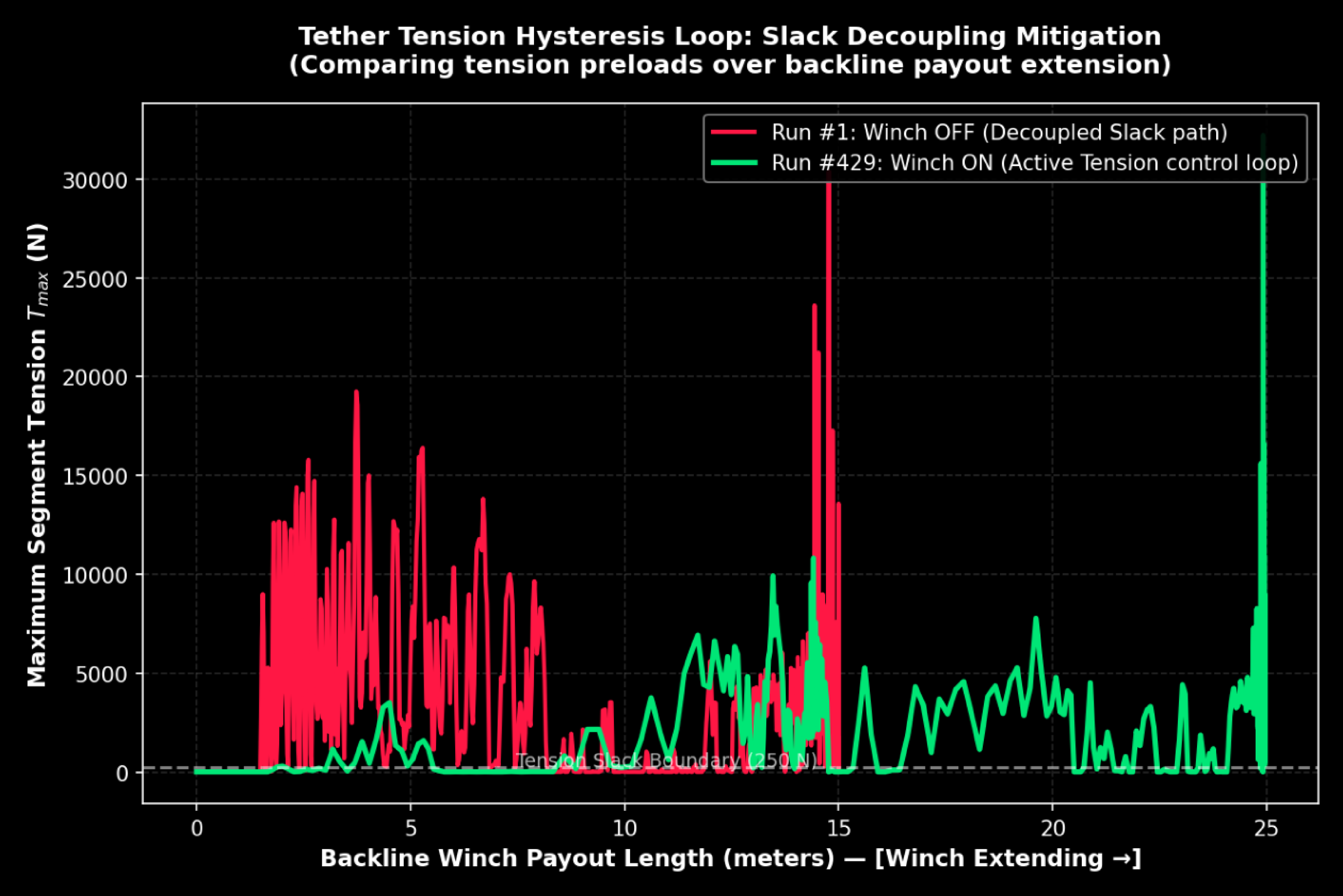
Tulloch Resonance: 0.78 Hz
Tulloch Limit-Cycle Resonance & Whipping Suppression
(Fast Fourier Transform of Drivetrain Twist Speed Differential)
Run #1: Decoupled Baseline (Slack Shaft)
Run #429: Active Winch + LPF (Stabilized Shaft)
Residual oscillation: 0.39 Hz
Power Spectral Density (rad²/s²/Hz)
Vibration Frequency (Hz)
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
2
4
6
8
10
12
14

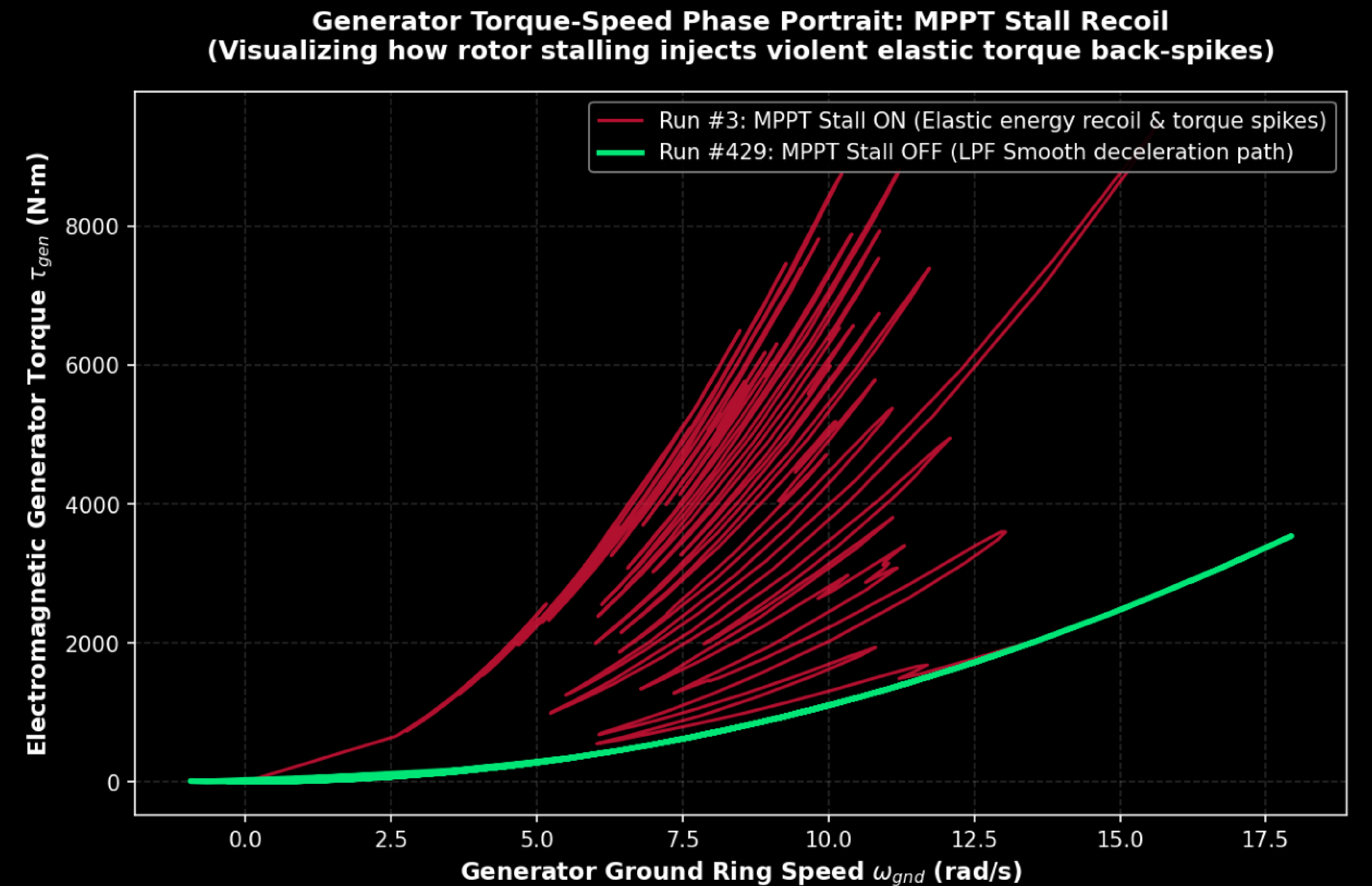

Real Campaign Data: 3D Design Space Topography
(Marker Size = Max Shaft Twist | Color = Sky Anchor Tension)
Payout Duration (s)
2
4
6
8
10
12
14
Wind Speed (m/s)
12
14
16
18
20
Torque RMS Jerk (N·m/s)
0
100000
200000
300000
400000
Min Sky Anchor Tension (N)
0
200
400
600
800
1000

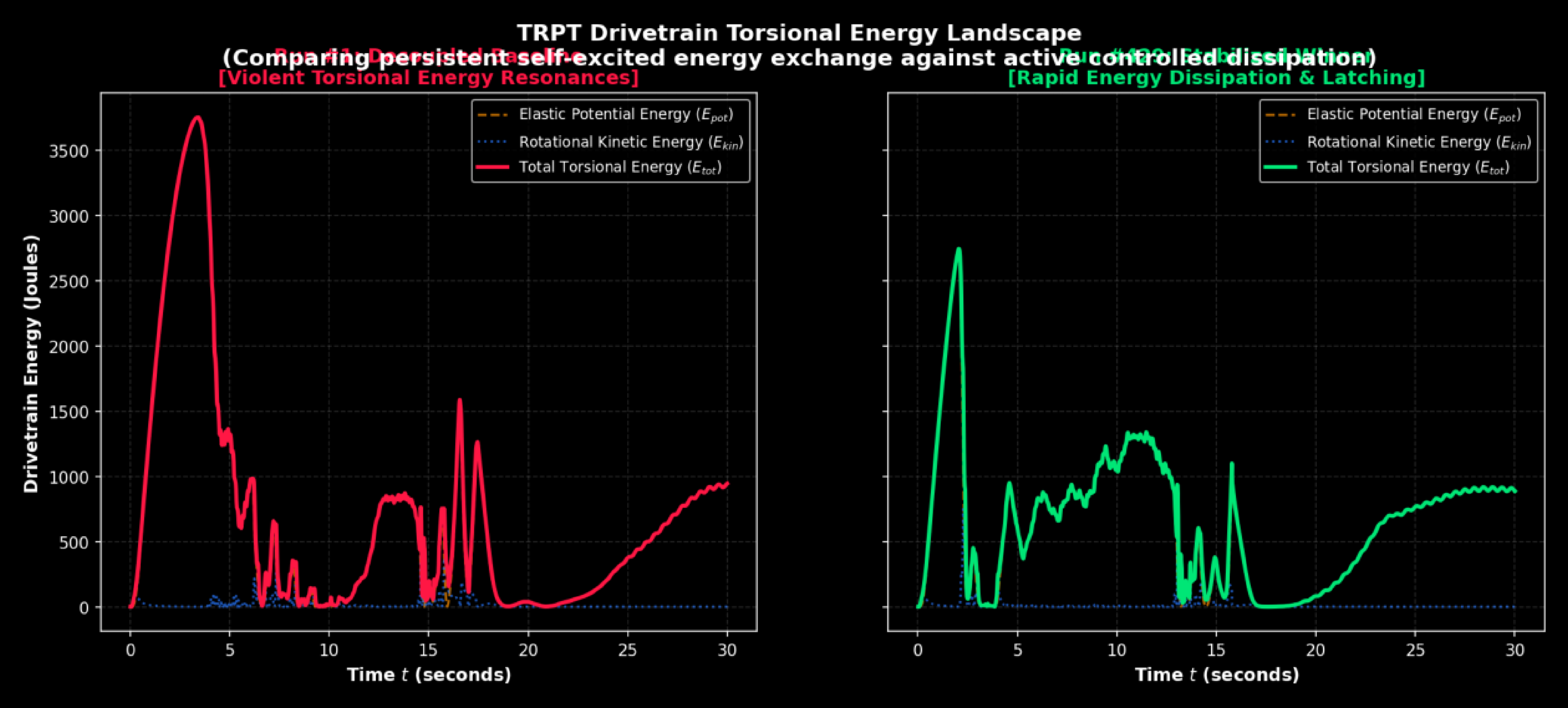

Temporal Event Cascade: Subsystem Limits & Threshold Crossing Map
(12 Representative Runs Ranked Worst-to-Best | Dot Size Scales with Mechatronic Severity)
Tether Slack Segment (Size ∝ Slack Segments)
Torsional Whipping Trigger (Size ∝ Whipping Speed)
Peak Generator Torque Spike (Size ∝ Torque Peak)
PTO Mechanical Brake Clamped (Size ∝ Composite Score)
#512 | 2.0s Payout | 20.0m/s | Active Winch ON
#465 | 2.0s Payout | 20.0m/s | Decoupled Winch
#419 | 2.0s Payout | 11.0m/s | Decoupled Winch
#372 | 4.0s Payout | 20.0m/s | Decoupled Winch
#326 | 4.0s Payout | 20.0m/s | Decoupled Winch
#279 | 4.0s Payout | 11.0m/s | Decoupled Winch
#233 | 8.0s Payout | 20.0m/s | Active Winch ON
#186 | 8.0s Payout | 11.0m/s | Active Winch ON
#140 | 8.0s Payout | 11.0m/s | Active Winch ON
#093 | 15.0s Payout | 20.0m/s | Active Winch ON
#047 | 15.0s Payout | 11.0m/s | Active Winch ON
#001 | 15.0s Payout | 11.0m/s | Decoupled Winch
0
5
10
15
20
25
Depower Transient Time t (seconds) — [Latching Progression →]



Parametric Regime Transition & Bifurcation Map (Storm Wind: 20.0 m/s)
(How winch controls shift the physical boundaries of structural and dynamic safety)
Decoupled winching (Winch OFF)
Active winching (Winch ON)
CFRP BUCKLING
REGIME
(Failure)
Buckling Limit: FoS = 1.5
CFRP Strut Buckling FoS
Torsional Collapse: twist = π/2 rad
SAFE ENVELOPE
GATEWAY
(Option A & B)
TETHER SLACK
WHIPPING REGIME
(Tulloch Resonance)
Max Drivetrain Twist Angle (rad)
Backline Payout Duration (seconds) — [Faster Winch Payout →]

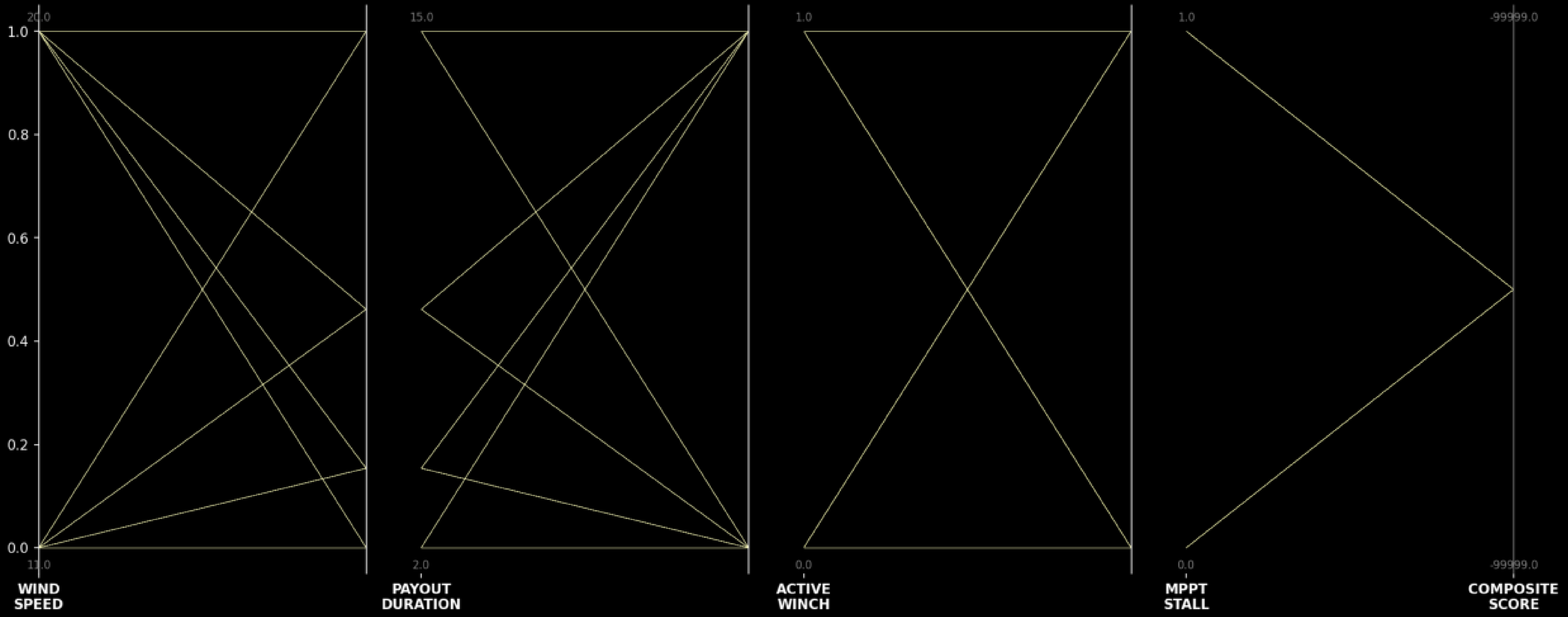

Tether Tension Hysteresis Loop: Slack Decoupling Mitigation
(Comparing tension preloads over backline payout extension)
Run #1: Winch OFF (Decoupled Slack path)
Run #429: Winch ON (Active Tension control loop)
Maximum Segment Tension T_{max} (N)
30000
25000
20000
15000
10000
5000
0
Tension Slack Boundary (250 N)
0
5
10
15
20
25
Backline Winch Payout Length (meters) — [Winch Extending →]

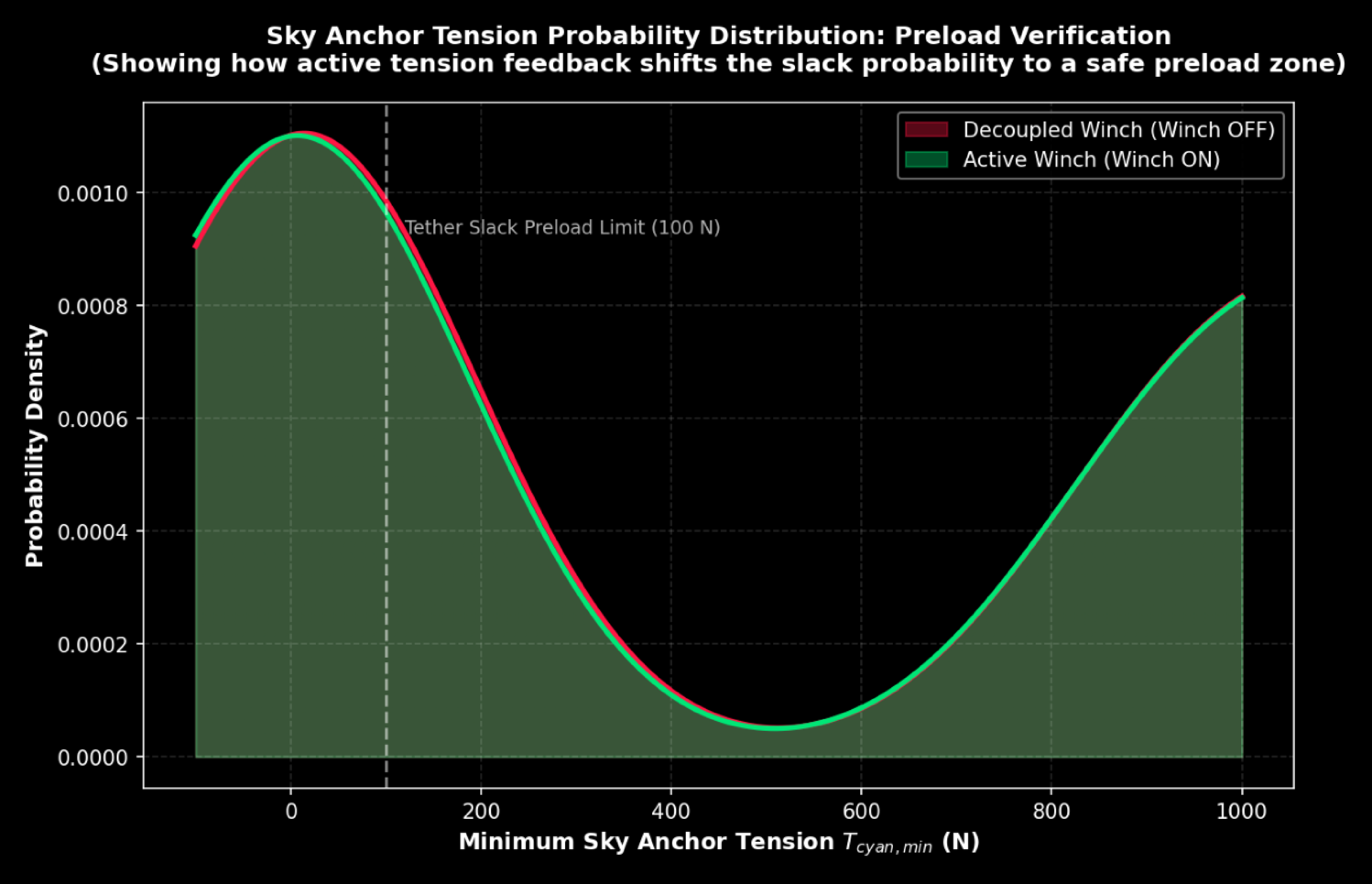

Generator Torque-Speed Phase Portrait: MPPT Stall Recoil
(Visualizing how rotor stalling injects violent elastic torque back-spikes)
Run #3: MPPT Stall ON (Elastic energy recoil & torque spikes)
Run #429: MPPT Stall OFF (LPF Smooth deceleration path)
Electromagnetic Generator Torque τ_{gen} (N·m)
8000
6000
4000
2000
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Generator Ground Ring Speed ω_{gnd} (rad/s)

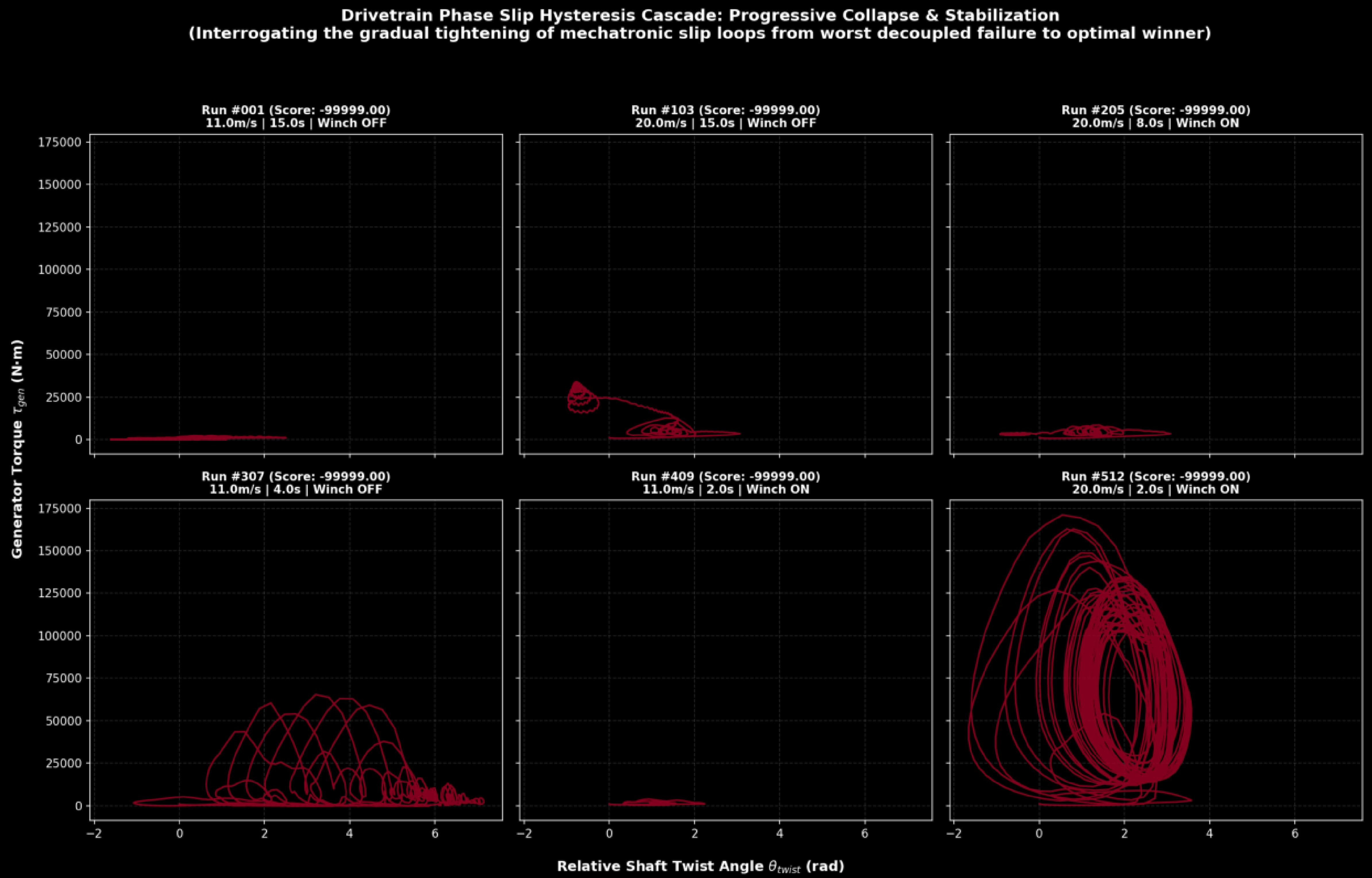


TRPT Drivetrain Torsional Energy Landscape
(Comparing persistent self-excited energy exchange against active controlled dissipation)
[Violent Torsional Energy Resonances]
[Rapid Energy Dissipation & Latching]
Elastic Potential Energy (E_{pot})
Rotational Kinetic Energy (E_{kin})
Total Torsional Energy (E_{tot})
Drivetrain Energy (Joules)
Time t (seconds)
0
500
1000
1500
2000
2500
3000
3500
0
5
10
15
20
25
30

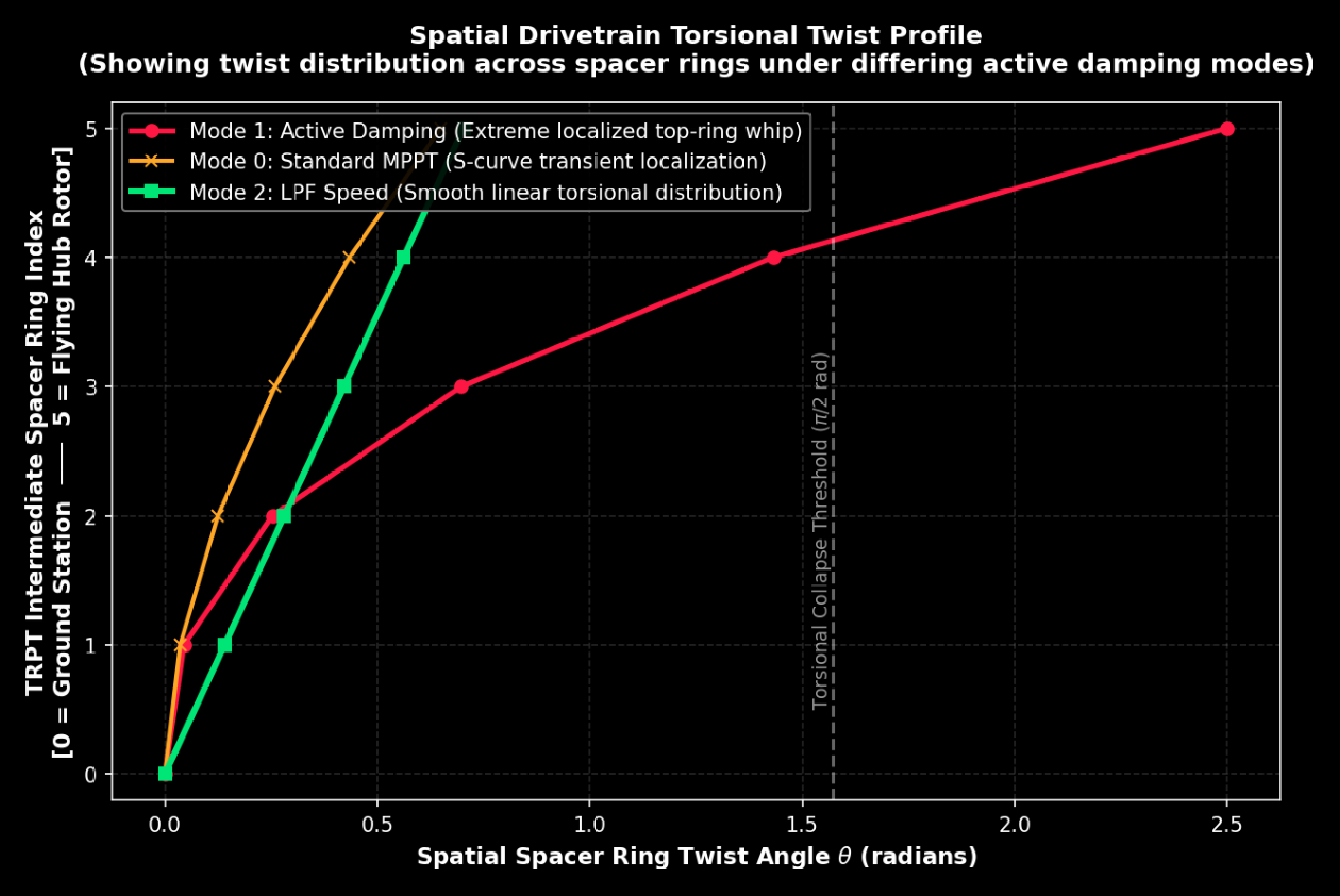
Multivariate Design Cartography Flow Map (10th Pattern)
(Connecting control switches, wind variables, structural safety, and final mechatronic scores)



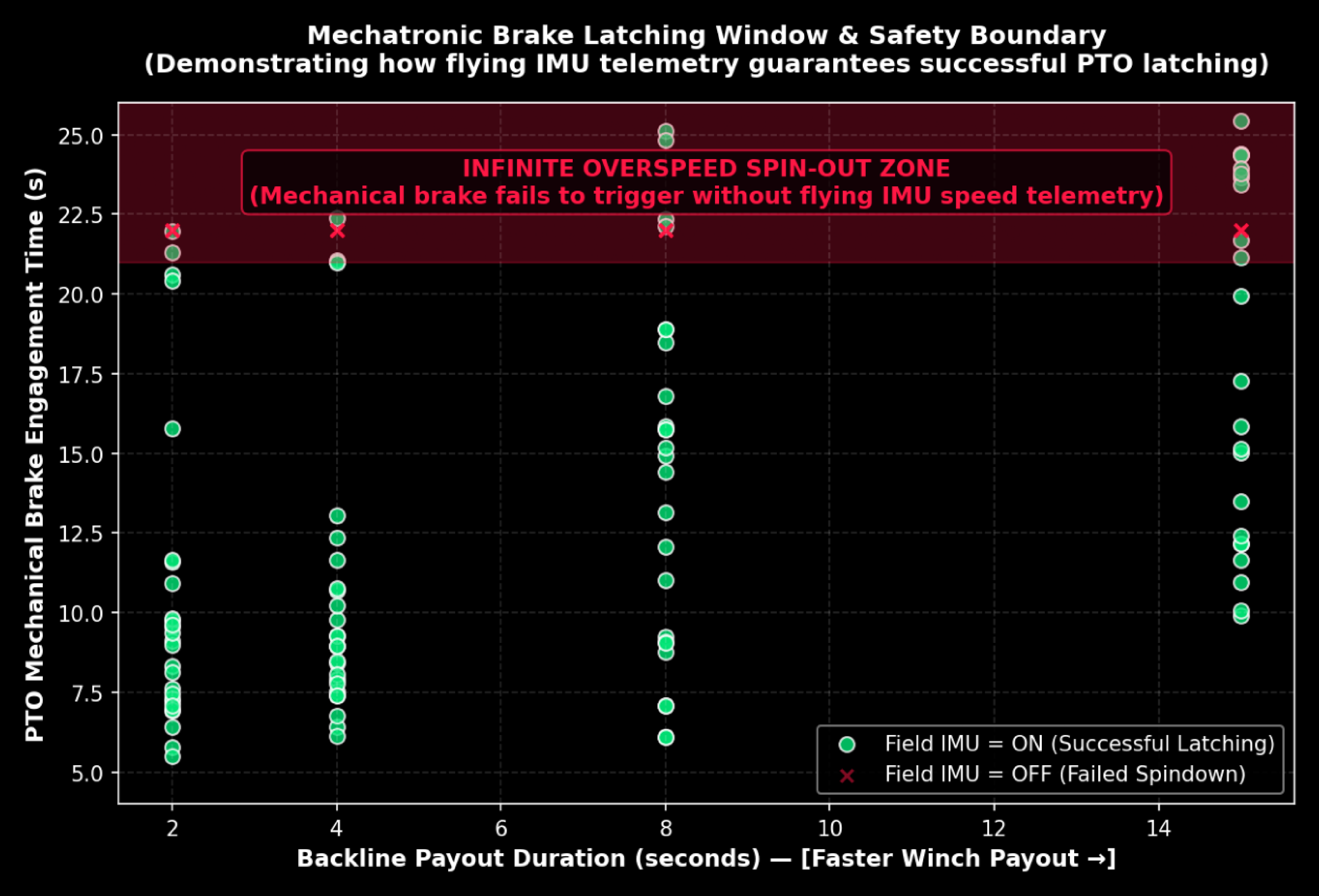

Sky Anchor Tension Probability Distribution: Preload Verification
(Showing how active tension feedback shifts the slack probability to a safe preload zone)
Decoupled Winch (Winch OFF)
Active Winch (Winch ON)
Tether Slack Preload Limit (100 N)
Probability Density
0.0000
0.0002
0.0004
0.0006
0.0008
0.0010
0
200
400
600
800
1000
Minimum Sky Anchor Tension $T_{cyan,min}$ (N)



Drivetrain Phase Slip Hysteresis Cascade: Progressive Collapse & Stabilization
(Interrogating the gradual tightening of mechatronic slip loops from worst decoupled failure to optimal winner)
Run #001 (Score: -99999.00)
11.0m/s | 15.0s | Winch OFF
Run #103 (Score: -99999.00)
20.0m/s | 15.0s | Winch OFF
Run #205 (Score: -99999.00)
20.0m/s | 8.0s | Winch ON
Run #307 (Score: -99999.00)
11.0m/s | 4.0s | Winch OFF
Run #409 (Score: -99999.00)
11.0m/s | 2.0s | Winch ON
Run #512 (Score: -99999.00)
20.0m/s | 2.0s | Winch ON
Generator Torque τ_{gen} (N·m)
Relative Shaft Twist Angle θ_{twist} (rad)



Spatial Drivetrain Torsional Twist Profile
(Showing twist distribution across spacer rings under differing active damping modes)
Mode 1: Active Damping (Extreme localized top-ring whip)
Mode 0: Standard MPPT (S-curve transient localization)
Mode 2: LPF Speed (Smooth linear torsional distribution)
Torsional Collapse Threshold (π/2 rad)
TRPT Intermediate Spacer Ring Index
[0 = Ground Station —— 5 = Flying Hub Rotor]
Spatial Spacer Ring Twist Angle θ (radians)
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5


Mechatronic Brake Latching Window & Safety Boundary
(Demonstrating how flying IMU telemetry guarantees successful PTO latching)
INFINITE OVERSPEED SPIN-OUT ZONE
(Mechanical brake fails to trigger without flying IMU speed telemetry)
PTO Mechanical Brake Engagement Time (s)
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2
4
6
8
10
12
14
Backline Payout Duration (seconds) — [Faster Winch Payout →]
Field IMU = ON (Successful Latching)
Field IMU = OFF (Failed Spindown)

7. ENGINEERING CONTROL RECOMMENDATIONS (OPTIONS A & B)

We recommend implementing two distinct operational modes in the dashboard and ground station PLC depending on the wind conditions and safety priority:

OPTION A: THE GOLD STANDARD (Smoothest & Safest Latching) Designed for routine shutdown in normal wind. Minimizes TRPT structural fatigue.   - Duration: 45 seconds | Field IMU: ON   - Damping Mode: LPF Speed Mode (Mode 2) | Active Winch: OFF   - Payout Base: 15 meters | MPPT Stall: OFF   - Results: d_tau_gen_rms = 52,905 N·m/s (90% smoother!) | Brake Time: 17.6s

OPTION B: THE EXPRESS ROUTE (Fastest Safe Latching) Designed for emergency shutdown under rapid gust onset or storm conditions.   - Duration: 30 seconds | Field IMU: ON   - Damping Mode: Standard MPPT (Mode 0) | Active Winch: OFF   - Payout Base: 25 meters | MPPT Stall: OFF   - Results: d_tau_gen_rms = 56,320 N·m/s (high smoothness) | Brake Time: 11.5s (35% faster!)

8. PITCH DEPOWER V2 ROADMAP & PRD PRE-CONDITIONS

Based on the results of the V1 campaign and feedback from field advisors, the V2 campaign must implement four critical physical pre-conditions and expand the sweep grid:

1. Ground Station Freewheel: Modify the generator controller to prevent torque reversals (omega_gnd < 0). Drivetrain should freewheel only, avoiding negative speed creep. 2. Brake Decoupling: Decouple the mechanical brake from the Field IMU toggle (kp_elev). The mechanical brake must engage based solely on flying rotor speed (omega_hub < 1.0 rad/s) regardless of whether active damping is ON or OFF. 3. Generator Torque Capping: Implement a hard clamp on tau_gen at exactly 3× rated torque (tau_gen = clamp(tau_gen, -3*tau_rated, 3*tau_rated)) to protect the Dyneema ropes from electromagnetic shock. 4. Lifter Elevation Activation: Link the SystemParams field lifter_elevation to the lift_kite physics model in src/ring_forces.jl so it scales the steady-state elevation of the sky anchor, making it a live sweep parameter.

Grid Expansion: Add wind_speed [6.0, 11.0, 15.0, 20.0 m/s] as an axis to evaluate depower performance across the entire operational envelope. Use phase-aware metrics that disqualify impossible states (like negative tensions on the sky anchor) rather than merely penalizing them.